
SMART ONESHOT
PHASE 3
NEET | ZOOLOGY
HUMAN HEALTH & DISEASES
EVOLUTION
THE LAST PHASE
MD SIR
SANKALP NEET
VEDANTU
Vedantu

Immunity

Innate
Non specific

External

Physical barrier → Skin, Mucus Epithe.

Physiological barrier → $\underbrace{\text{Mucus, Saliva, Tears}}_{\text{Lysozymes}}$, $\underbrace{\text{HCl}}_{\text{Stomach}}$

Cellular barrier
- Non Phagocytic Cells → Natural Killer Cells — Kill
  - Cancer cells
  - Virus infected cells
- Phagocytic Cells
  - Blood = Monocytes, Nutrophils (PMNL)
  - Conn. Tissue = Macrophage

Cytokine barrier → Virus infected cells Release Protein → $\boxed{\text{Interferon}}$

Internal

Interferon protect Non infected cells from Viral infection

Acquired
Specific

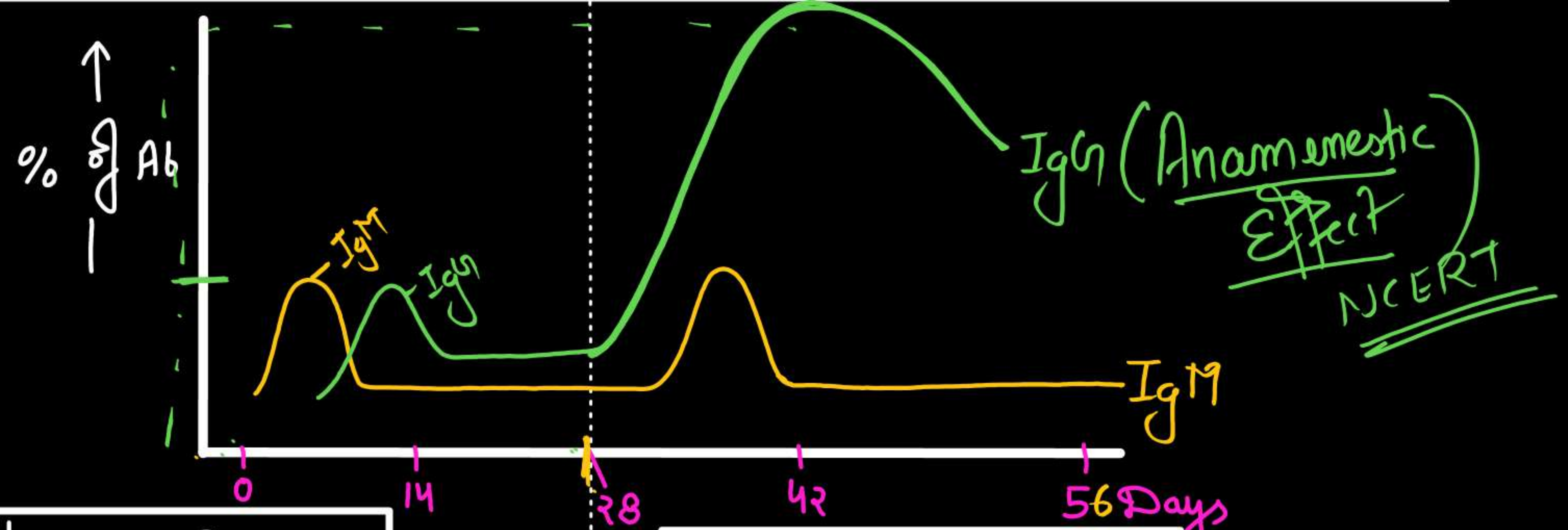
ACQUIRED IMMUNITY-

Acquired immunity, on the other hand is $\boxed{\text{pathogen specific.}}$ It is characterised by $\boxed{\text{memory.}}$ @

Features of Acquired Immunity –

I. Specificity ✓
II. Diversity ✓
III. Memory ✓
IV. $\boxed{\text{Discrimination}}$ b/w

- Self = No Response.
- Non Self = Responses
  - Pri. Immune Responses.
  - Sec. Immune Responses



Ist Immun Response

- Ist Encounter.
- Involve - B-lym.
- Formation of Ab = Low Intencity
- IgM > IgG

IInd Immun Response

- IInd Encounter - Same antigen
- Involve Memory cells.
- High Intencity
- IgG >>>> IgM

Structure of Antibody (Ab)- (Immunoglobulin) = Glycoprotein

4-Polypeptides ($H_2 L_2$)

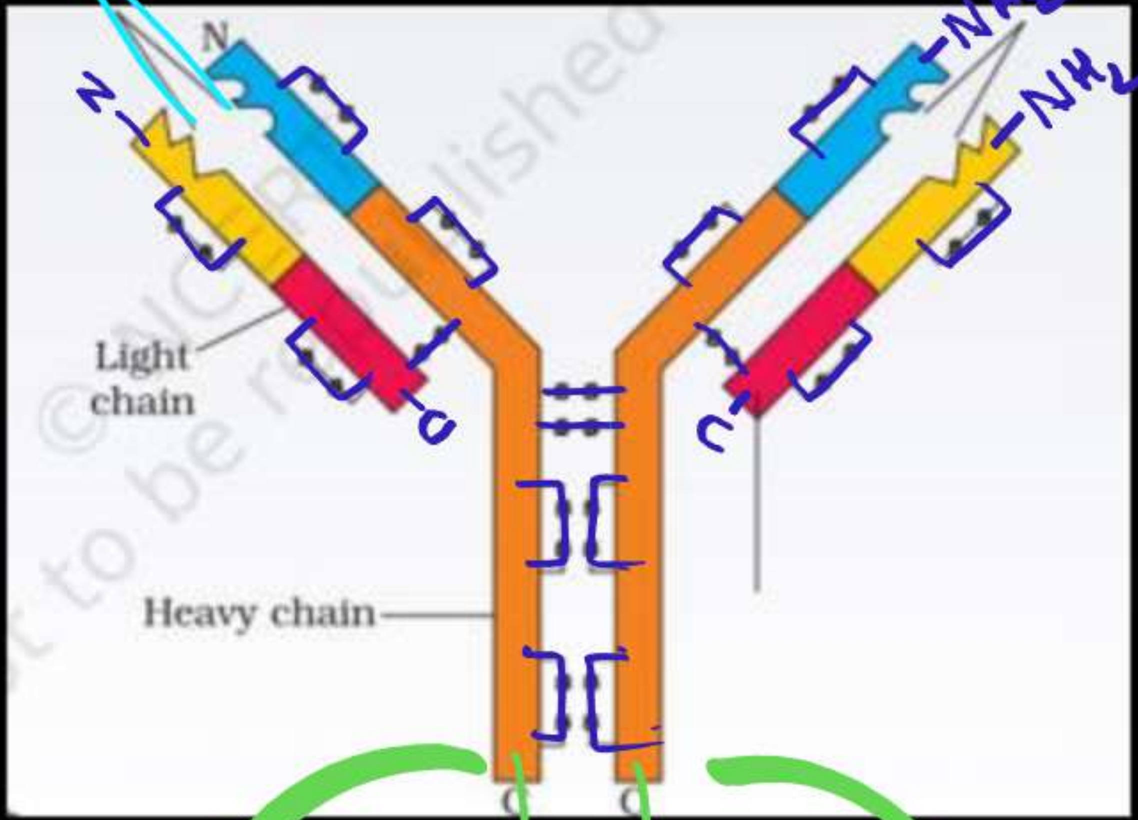
Two Heavy chain — Two light chain

Antigen binding Site

Total No. of Disulphide bond = 16

b/w Heavy chain = ?

b/w H- Light chain = ?



HOOC COOH

γ - IgG (Max^m) 80%

μ - IgM

α - IgA

δ - IgD

ε - IgE (Min^m) <1%

Types of Acquired Immunity

(B-lymphocytes based)
or
Antibody Mediated Imm. (AMI)
or
Humoral Immunity
(Body fluid)

Active B-lym.
[Plasma cells]
↓
(Formation of Antibody)

Inactive B-lym.
[Memory cells]

(T-lymphocytes based)
or
Cell Mediated Immunity (CMI)
↓

Responsible for Tissue grafting or Organ Transplantation

HIV

- T_H ⟶ Control all Immun cells.
- T_K ⟶ Kill — [Cancer cells, Virus Infected cells]
- T_S
- T_{Memory} ✓

Tissue graft/ Organ transplantation (Cell Mediated Immunity)

- Matching
  - → Blood Group
  - → Human leucocyte Antigen (HLA)
    - → MHC-I
    - → MHC-II
- Immuno suppresive Drugs
  - Cyclosporin
  - Cortisol

Active and Passive Immunity

Antibody Produced by Host	Prepaired antibody Provided to Host
A	P

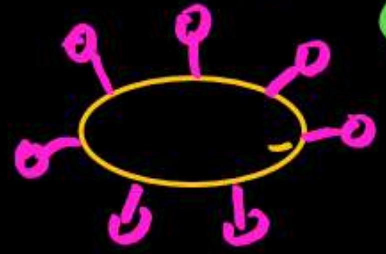
A
- Natural
  - All Infection
- Artificial
  - All Vaccination

P
- Natural
  - IgA = Colostrum
  - IgG = Placenta
- Artificial
  - A.T.S.
  - Antivenum

Vaccination

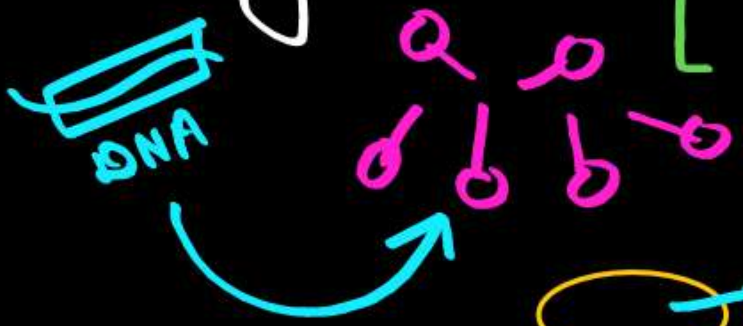
Imp

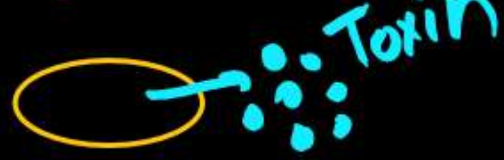
- Ist-generation [whole Organism]
  - Killed
  - Live attenuated (weak)
  - eg- OPV, BCG, Small pox etc.



- IInd-generation [Sub-unit]
  - Recombinant → Hepatitis-B
  - Toxoids [Inactive toxin] → D.P.T.

NCERT





- IIIrd-generation [DNA] → Hepatitis-B

Vaccines

Allergy

- Antibody Responsible for allergy = IgE
- Chemicals " " " " = Mast Cells — [Histamine, Serotonin]
- Anti allergic Drugs = Antihistamin, Adrenalin & Steroids.

Autoimmunity

= Due to - Genetic — Unknown Reason
- eg = Rheumatoid arthritis etc.

Types of lymphoid organs

- Pri. Lym. Organ { Formation & Maturation } Red bone marrow & Thymus
- Sec. lym. Organ { Interaction of lympho. } Spleen, Lymph Nodes, MALT etc.
  - MALT → 50%

HEALTH

According to WHO - health a state of complete-

- Physical
- Mental
- Social well being.

DISEASES

- Any Change from Normal State that caused discomfort and affect the working of Organs and system, characterised by – Some signs & Symptoms

Types of diseases

Congenital diseases

(Genetical Disease)

Acquired diseases

Communicable (Infectious)

Spread through —
- Bacteria
- Virus
- Fungus
- Helminthes.
- Protozoa etc.

eg - AIDS

Noncomunicable (Non infectious)

eg
- Diabetes.
- Cancer.
- Deficiency Disorders
- Drugs & Alcohol Abuse

= etc.

Bacterial Diseases	Pathogen	Symptoms & Pathogenicity
Typhoid	*Salmonella typhi*	➢ Enters small intestine through **contaminated food and water** and migrate to other organs through blood. ➢ Sustained high **fever (39-40°C).** ➢ Weakness, constipation, stomach pain, Headache, loss of appetite. ➢ In severe cases **intestinal perforation, (death).** ➢ Confirmed by **widal test**. ➢ Mary Mallon, **nicknamed-Typhoid Mary** (Carrier of typhoid)
Pneumonia	*Streptococcus pneumoniae* *Haemophilus influenzae*	➢ By **droplet or aerosol infection** ➢ Infects alveoli of the lungs. ➢ Alveoli get filled with fluid leading to severe problem in respiration. ➢ Fever with chills, cough & headache. ➢ In **severe cases lips and nails turns gray to bluish in colour**.

Diphtheria	**Corynebacterium diphtheriae**	➢ **High grade fever, affects throat.** ➢ **Causes suffocation**
Plague (Black death)	Yersinia pestis (Parasite of - Rat flea/ Xenopsylla cheopis)	➢ High fever, headache. ➢ Enlargement of axillary lymph nodes ➢ Red patches appear on skin which turns black - Death

Viral Diseases	Pathogen	Symptoms & Pathogenicity
Common cold	Rhino viruses (Group of viruses)	➢ One of the **most infectious** human ailments. ➢ Transmits through **droplet** resulting from cough, sneeze etc. ➢ Infect nose and respiratory passage **but not the lungs**. ➢ Nasal congestion and discharge, sore throat, hoarseness, cough, headache, tiredness. ➢ Usually last **for 3-7 days.**
Hepatitis-B	HBV (ds DNA)	➢ Severe liver damage, jaundice. ➢ Recombination DNA vaccine ➢ Transmits-through parenteral and sexual route (Can cross placenta)
Polio	Polio virus (ss RNA)	➢ Inflammation of nervous system, stiffness of the neck, weakness of particular skeletal muscles. ➢ High fever, pain in body, paralysis of limb muscles.

DENGUE

1) Causative agent = Dengue Virus – RNA

[Flaviviridae – family]

2) = Vector = Female Aedes aegypti Mosquito

CHIKUNGUNYA

Causative agent = Chikungunya Virus – RNA

[Togaviridae – family]

2) – Vector = Female Aedes aegypti Mosquito

Helminthic Diseases	Pathogen	Symptoms & Pathogenicity
Ascariasis	Ascaris Lumbricoides (Common Round worm)	➢ Intestinal parasite ➢ Internal bleeding, muscular pain, fever, anaemia, blockage of the intestinal passage. ➢ Eggs of the parasite excreted along with the faeces of infected person which contaminate soil, water, plants etc. ➢ Infection occurs by taking contaminated water, fruits, vegetables etc.
Elephantiasis (Filariasis)	Wuchereria (W. bancrofti) (Filarial worm)	➢ Slowly developing chronic inflammation of the organs in which they live for many years. ➢ Usually the lymphatic vessels of the lower limbs get blocked. ➢ Genital organs are also often affected, (swelling in tests & scrotum). ➢ Transmitted to a healthy person through the bite by the female mosquito vector (Mainly Culex).

Fungal Diseases	Pathogen	Symptoms & Pathogenicity
Ringworms	Microsporum, Trichophyton & Epidermophyton	➢ One of the most common infectious disease. ➢ Appearance of dry, scaly lesions on various parts of the body such as skin, nails and scalp. ➢ Generally acquired from soil or by using towels, clothes or even the comb of infected individuals.

Figure 8.3 Diagram showing ringworm affected area of the skin

Protozoan Diseases

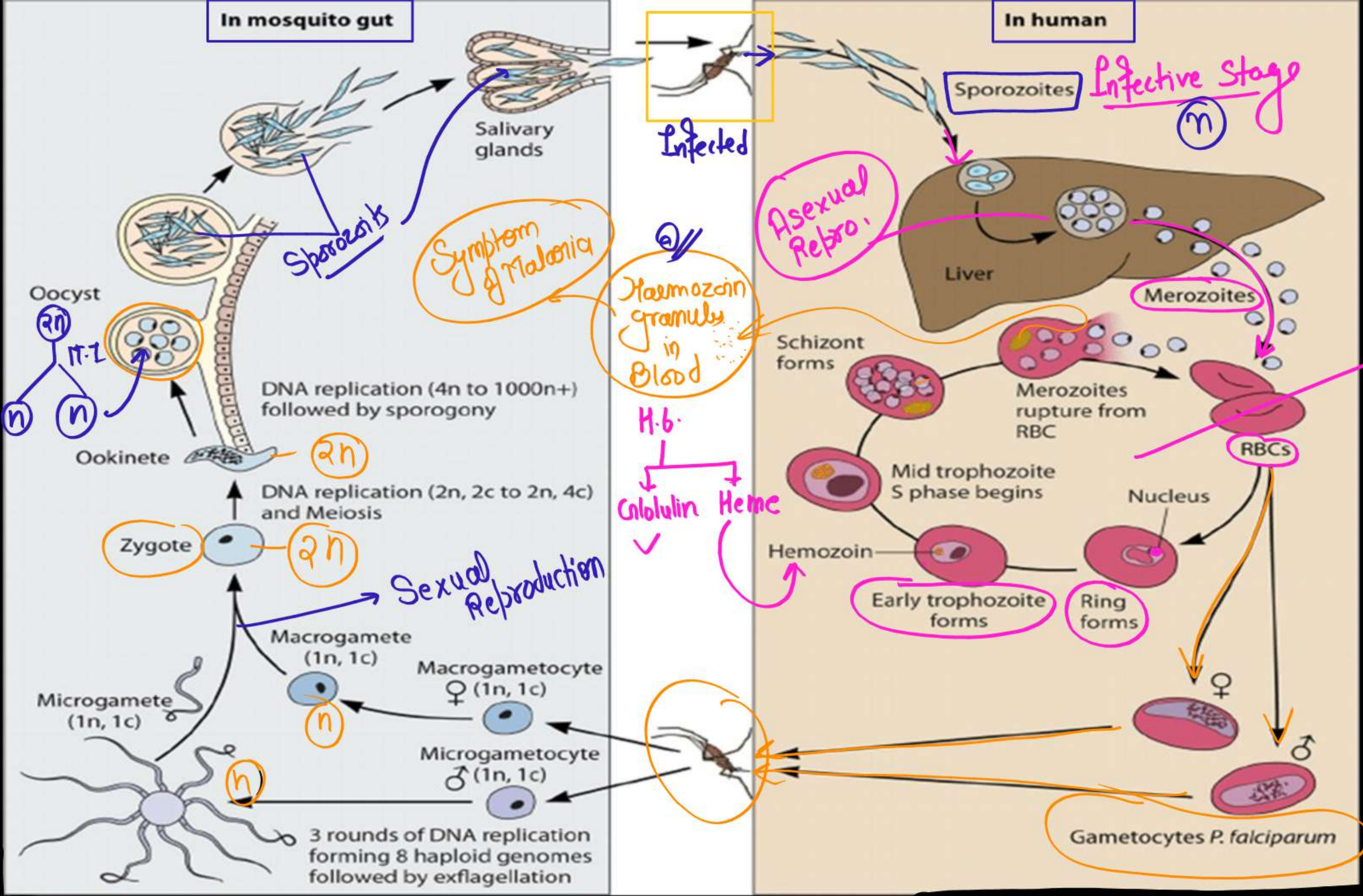
① **Malaria** → Protozoa (Plasmodium) @

Different species of $\boxed{\text{Plasmodium}}$ which infect humans are-

- P. Vivax [60-65%] → @ $\boxed{\text{Benign}}$ Tertian Malaria
- P. Malaria [1%] → Quartan Malaria.
- @ P. falciparum [30-40%] → $\boxed{\text{Malignant Tertian Malaria}}$ NCERT

Lifecycle → Digenetic → Pri. Host = Mosquito (Female Anaphilis) = Stomach @

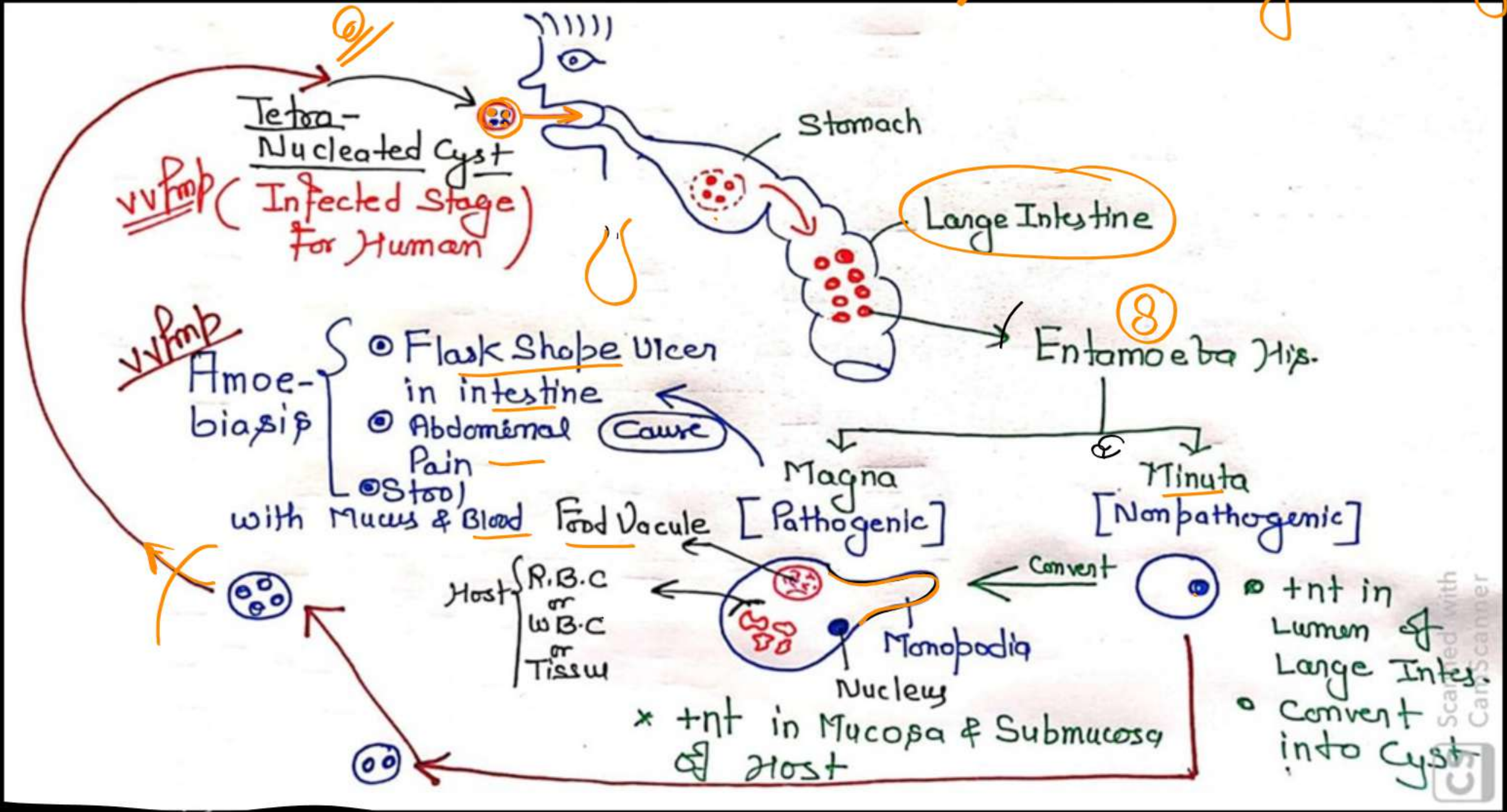
→ Sec. Host = Man → Liver, R.B.C


In mosquito gut
In human
Salivary glands
Infected
Sporozoites
Infective Stage
n
Sporozoits
Oocyst
2n
M.I
n
n
Symptom of Malaria
Haemozoin granules in Blood
Asexual Repro.
Liver
Merozoites
Schizont forms
Merozoites rupture from RBC
Erythrocytic Cycle
RBCs
DNA replication (4n to 1000n+) followed by sporogony
Ookinete
2n
H.B.
Globulin
Heme
Mid trophozoite S phase begins
Nucleus
DNA replication (2n, 2c to 2n, 4c) and Meiosis
Zygote
2n
Hemozoin
Early trophozoite forms
Ring forms
Sexual Reproduction
Macrogamete (1n, 1c)
Macrogametocyte ♀ (1n, 1c)
n
Microgamete (1n, 1c)
Microgametocyte ♂ (1n, 1c)
n
♀
♂
3 rounds of DNA replication forming 8 haploid genomes followed by exflagellation
Gametocytes P. falciparum

(2) Amoebiasis (amoebic dysentery)- Entamoeba histolytica

(parasite in the large intestine of human)

Monogenitic life cycle

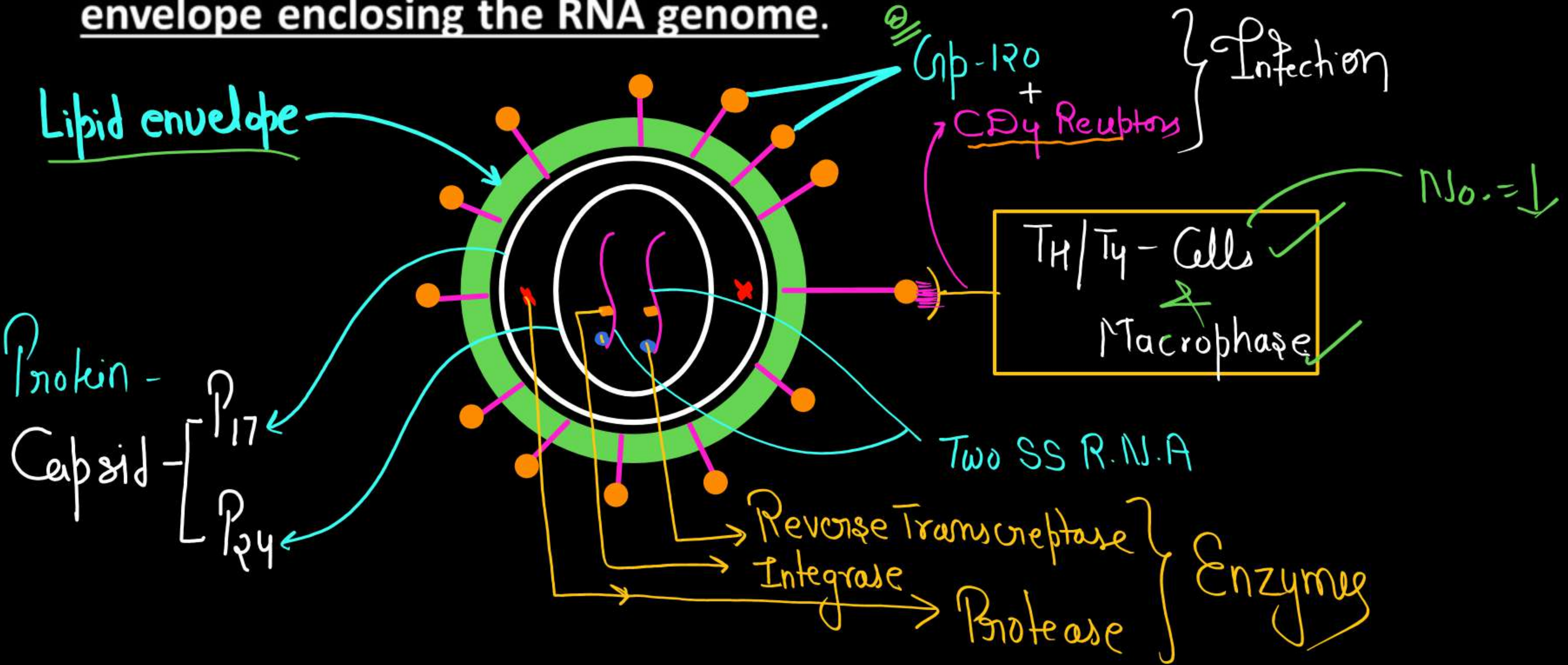


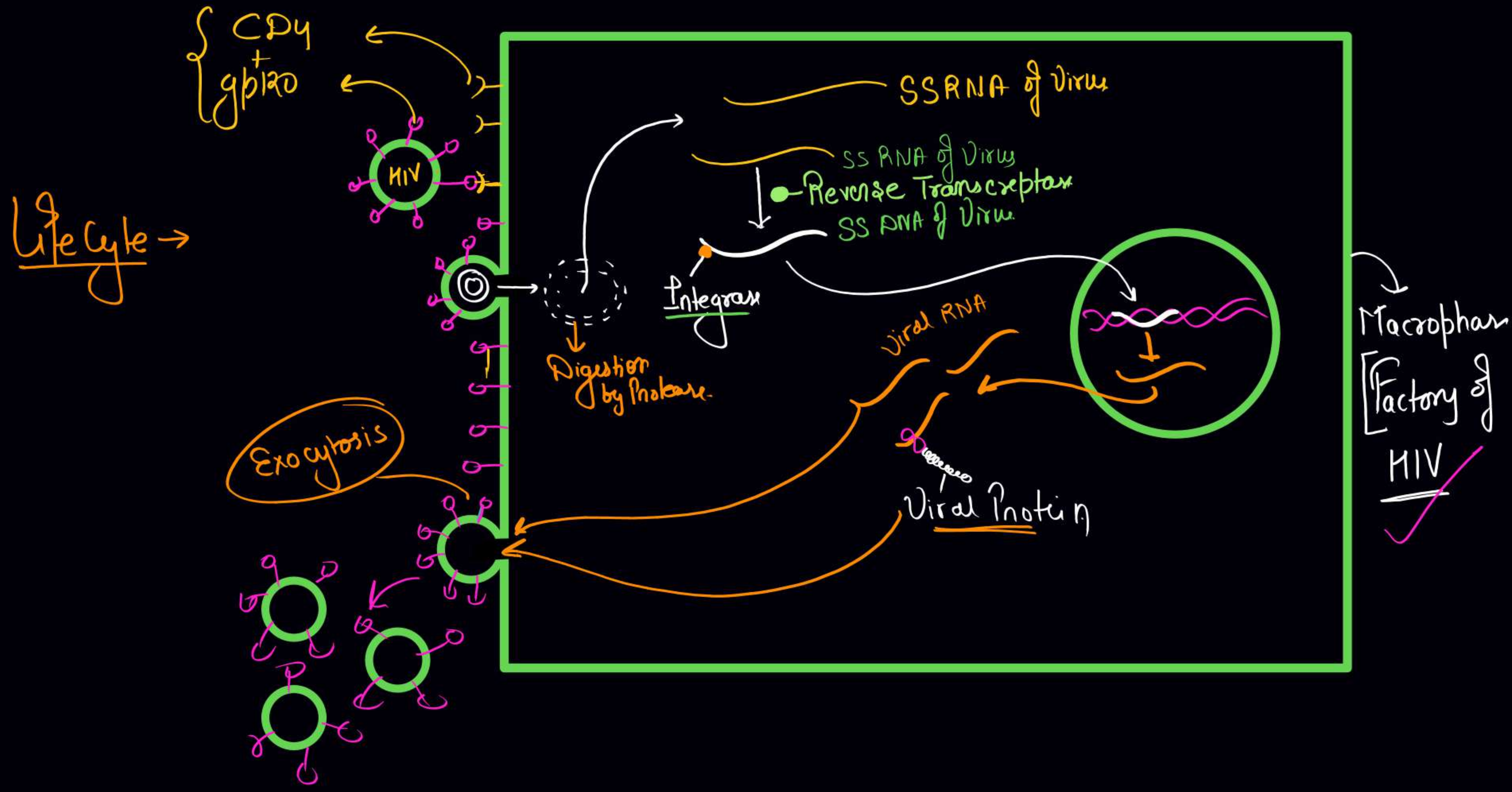
AIDS

- The word AIDS stands for **Acquired Immuno Deficiency Syndrome.** This means deficiency of immune system, acquired during the lifetime of an individual indicating that it is not a congenital disease.
- 'Syndrome' means a group of symptoms.
- AIDS was first reported in **1981** and in the last twenty-five years or so, it has spread all over the world killing more than **25 million persons**

HIV

- AIDS is caused by the Human Immuno deficiency Virus (HIV), a member of a group of viruses called **retrovirus,** which have an **envelope enclosing the RNA genome**.




CD4
+
gp120
HIV
Life Cycle →
SSRNA of Virus
SS RNA of Virus
Reverse Transcriptase
SS DNA of Virus
Integrase
Digestion by Protease
Viral RNA
Viral Protein
Exocytosis
Macrophage
[Factory of HIV]

Diagnostic test and Treatment

- A widely used diagnostic test for AIDS is - Enzyme Linked Immuno-Sorbent Assay (ELISA). Confirmation test – Western blotting

for Viral Protein

- Treatment of AIDS with anti-retroviral drugs is only partially effective.
- They can only prolong the life of the patient but cannot prevent death, which is inevitable.

[NACO]

Drugs →

1) - Reverse transcriptase inhibitor → A.Z.T

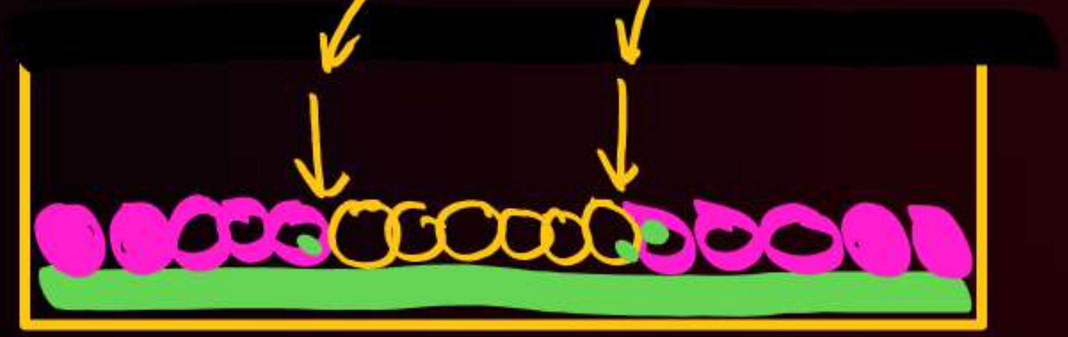
2) → Protease inhibitor → Ritonavire

Cancer

- Cancer is one of the most dreaded diseases of human beings and is a major cause of death all over the globe.
- More than a million Indians suffer from cancer and a large number of them die from it annually

Difference between Normal Cell and Cancer Cell

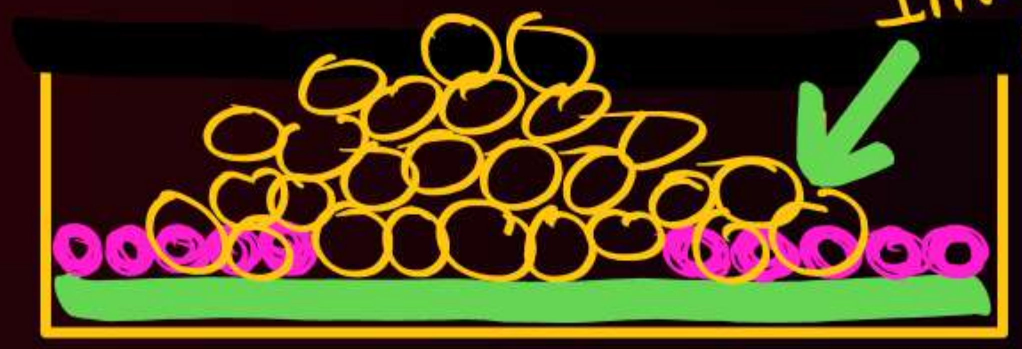
* [Control Cell Division]
* Contact inhibition +nt



* Invasiveness = −nt
* Mortality = +nt

· [Uncontrol Cell Division]

Invegination +nt

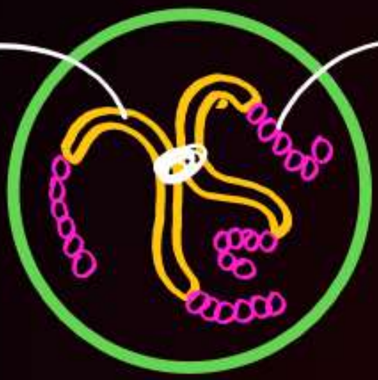


+nt

−nt Immortal

- Telomerase Enzyme ⇒ Inactive

Chromosom



Telomer Seq.
[Neucleotide]

Telomerase Inactive → 50 generation

After 50 generation

Teromer Seq. (−nt)

Program cell Death

⇓

(Apoptosis)

- Telomerase = Active

After 50 generation

Telomer Seq. −

Maintain by − Telomerase Enzyme.

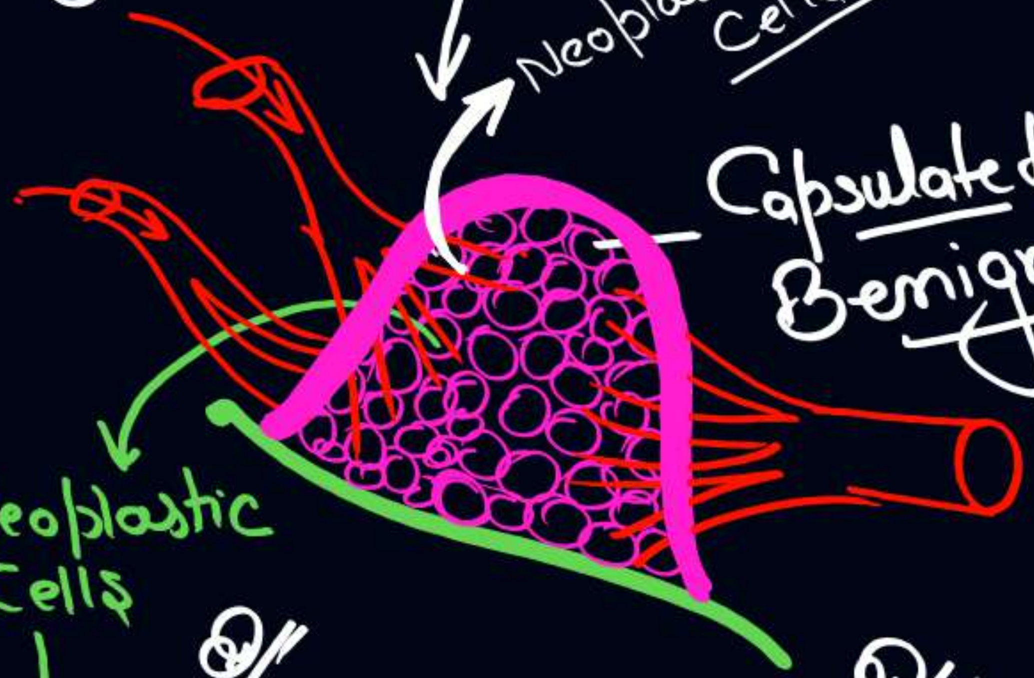
Apoptosis (−nt)

Chemical Carcinogen
or
Physical Carcino-gen
Biological Carcinogen

Normal cell

Onchogenic Transformation

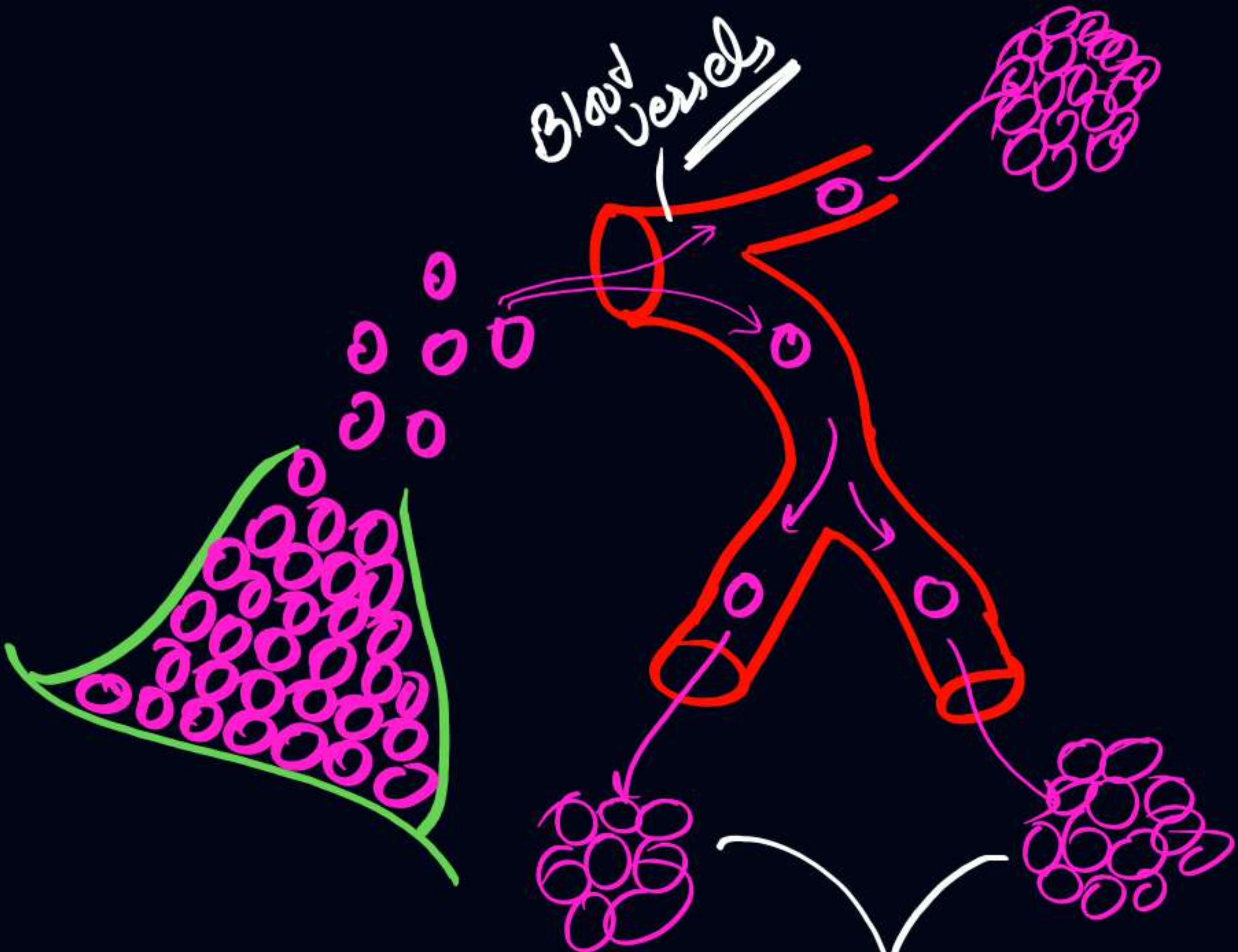
Neoplastic Cells



Capsulated Benign Tumor } Q// Neoplasm = Group of Neoplastic Cells

{ Neoplastic Cells

Release → Q// Angiogenasis Factors } → Q// ↑ No. of Blood Vessels around Tumor



Malignant Tumor
or
Cancer

vvvImp

{ Spreading of Tumor Cells } Metastasis

Causes of cancer :

Physical, chemical or biological agents. These agents are called carcinogens.

Physical carcinogens –

Ionising radiations - X-rays and gamma rays

Non-ionizing radiations - UV cause DNA damage

Chemical carcinogens-

Tobacco smoke have been identified as a major cause of lung cancer etc.

Biological agents –

Cancer causing viruses called oncogenic viruses like – HPV, HBV etc.

Cancer detection and diagnosis :

- Biopsy and histopathological studies
- Radiography (use of X-rays),
- CT (computed tomography) X-rays
- MRI (magnetic resonance imaging) – Safest technique Q
- Q Antibodies against cancer -specific antigens are also used for detection of certain cancers.
- Techniques of molecular biology can be applied to detect genes in individuals with inherited susceptibility to certain cancers.

Treatment of cancer :

The common approaches for treatment of cancer are

- Surgery
- Radiation therapy
- Chemotherapy – Vincristine and vinblastine
- Immunotherapy - α-interferon

Drugs

Psychotropic
(Mood altering)

eg-
- Tranquillizer
- Opiods
- Stimulants

Psychedelic
(Hallucinogenic)

eg-
- Cannabinoids
- Datura
- Atropa belladona
- L.S.D (Synthutic)

① Tranquillizer

- → Anesthetic ✓
- → Depressent ✓
- → Sleeping drugs ✓
- → ↑ - Tranquility

eg → ⓐ Benzodiazepine (Valum) → used in
- Anxiety
  ↓
- ⓐ Insomnia
- ⓐ Schizophrinia

\+

ⓑ Barbiturate

→ Mechanism
↓
↑ GABA in Brain

Opioids →

- → Narcotics ✓ नशीले
- → Sedative
- → Depressents
- → Analgesic (Pain Killer) ✓✓✓ Imp.
- → Euphoriant

Q.//

- → Isolate from → Papaver somniferum (opium poppy) Q// → (Latex)
- → Contain ⇒ 20 - Alkaloids ✓
- → Derivatives of Opium poppy →

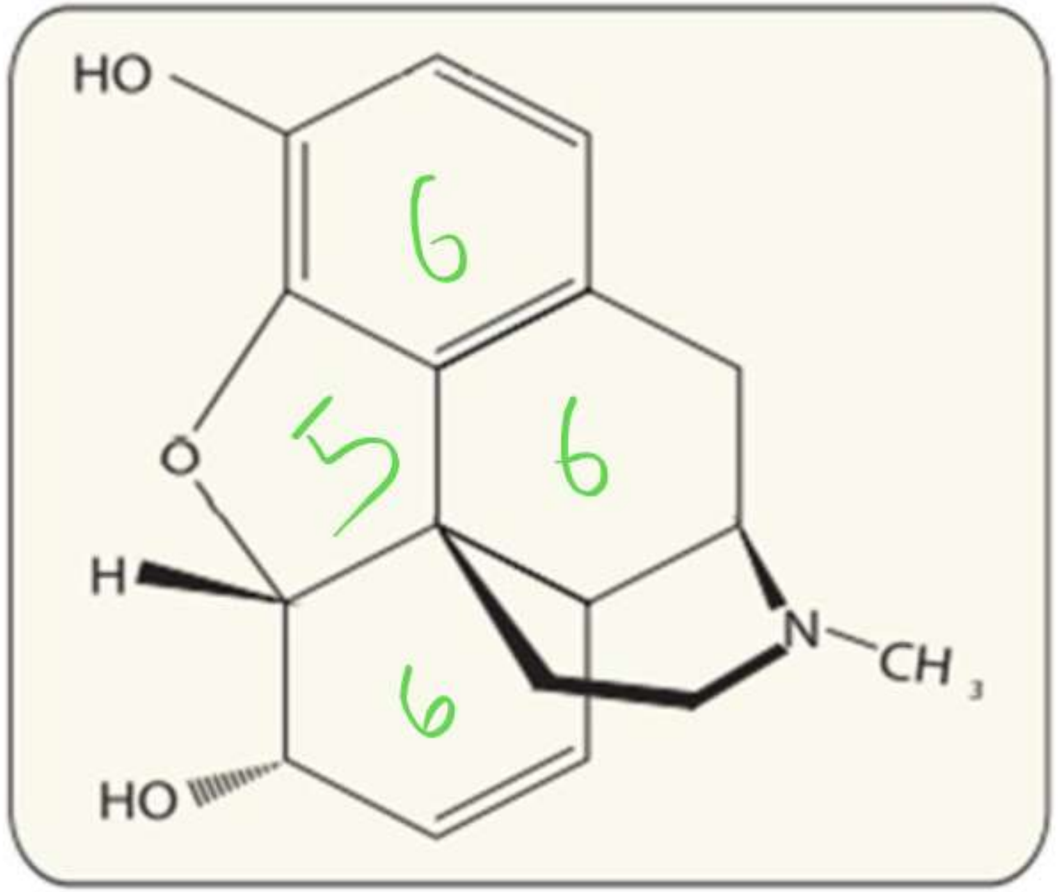


Figure 8.7 Chemical structure of Morphine

Figure 8.8 Opium poppy

* Morphine → • Analgesic (Pain killer)
  • Receptor t nt on — [Brain, Gut]

$\xrightarrow{\text{Methylation}}$

* Codein → • Methy-morphine
  • used in Cough Syrup

$\xrightarrow{\text{Acetylation}}$

* Heroine → • Diacetylation of morphine

* Smack → • Produced from Heroine
  • Brown sugar
  • Taken by ⟶ Snorting & Injection

↓
· White Crystals
· Bitter & odourless

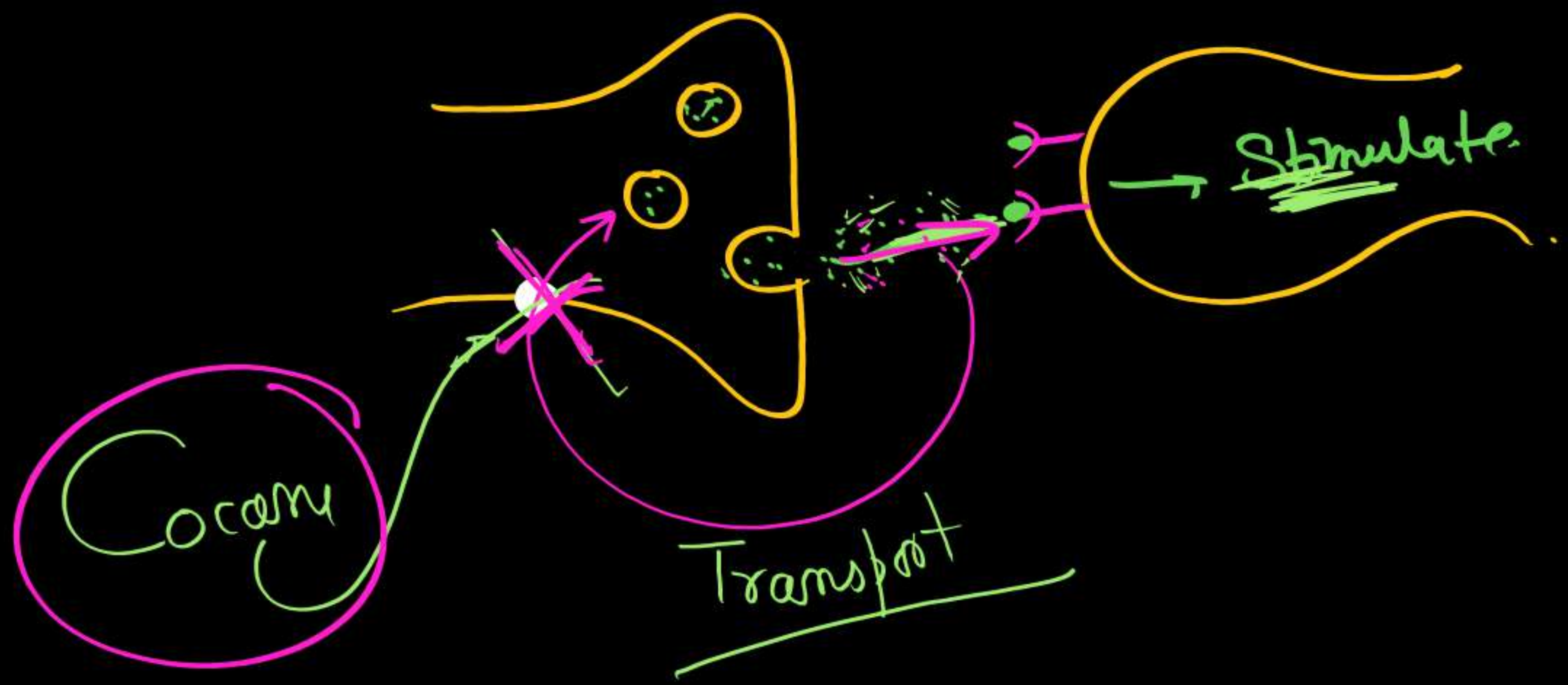
Stimulants

- Cocaine ✓
- Tobaco ✓
- Amphitamines ×

1) Cocaine
or
Coke/Crack

- → Isolate from = Coca plant (Erythroxylum Coca) @
  @ [Native to South America] @
- → Taken by = Snorting @
- → Euphoriant, ↑ - Energy
- → Excessive dosage ⇒ Hallucination @

* Stimulate → C.N.S., by interferes with transport of Dopamine



Tobaco

- Nicotine alkaloid +nt
- Taken by = Smoked, Chewed, Snuff
- Stimulate = Adrenal glands

↑ Ardenalin & Nor Adrenalin.

↑ - Glucose level

- Associated with ⇒ Mouth Cancer
- Smoking –
  - ↑ - CO in blood
  - Lung Cancer

Amphetamine →

- → Synthetic Drug
- → Supermen Drugs
- → Antisleep Drugs
- → Included in dope test

CANNABINOIDS (Hallucinogenic)

* Receptor prnt in ⇒ Brain @//
* Isolate from plant ⇒ Cannabis sativa @//

eg → Marijuana → [Active ingradient (Delta-9-T.H.C.)
[Dried - Flower @//

Bhang → Fresh leaves @//

Ganja → Female inflorescence @//

Charas or Hashish → Resinous Extract @//

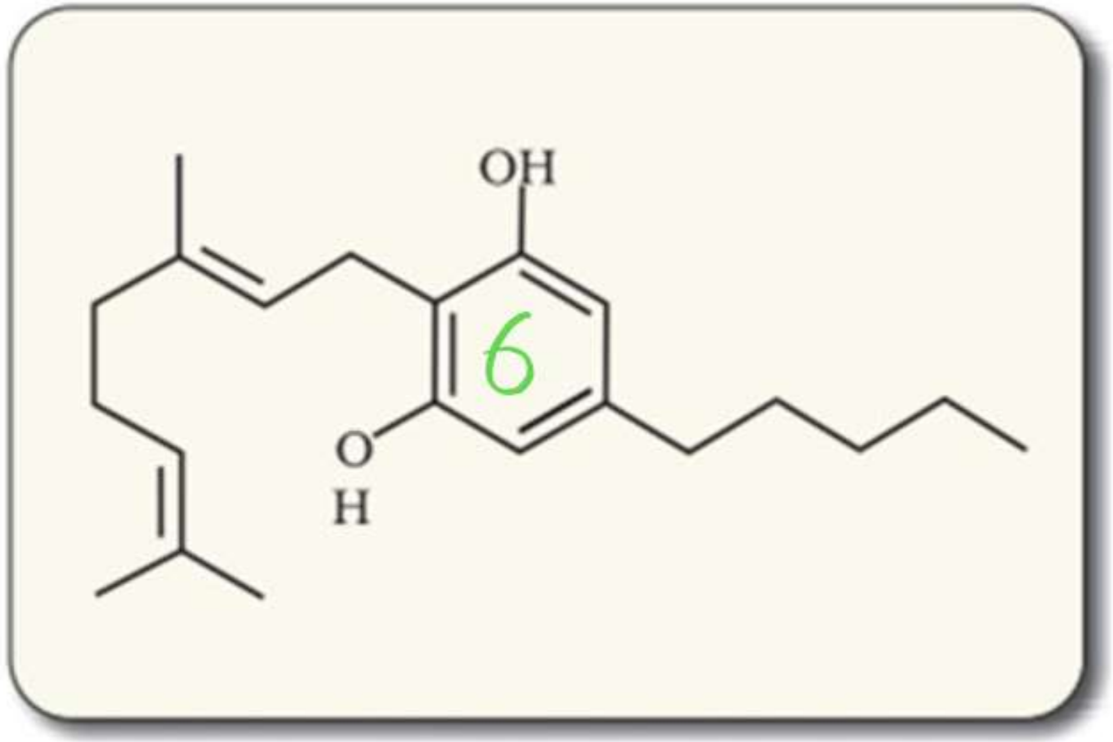


Figure 8.9 Skeletal structure of cannabinoid molecule

Figure 8.10 Leaves of *Cannabis sativa*

Lysergic acid diethylamide (L.S.D)

→ Synthetic Hallucinogen

→ obtain from = Ergot of fruting body of fungus ⇒ Claviceps purpurea   vvImp

Datura

&

Atropa belladona

Figure 8.11 Flowering branch of *Datura*

Neoplastic characteristics of cells refer to:

A. A mass of proliferating cell ✓

B. Rapid growth of cells ✓

C. Invasion and damage to the surrounding tissue ✓

D. Those confined to original location ✗

Choose the correct answer from the options given below (NEET 2025) ✓

(a) A, B, D only (b) B, C, D only

(c) A, B only (d) A. B, C only ✓

d

Which are correct:

A. Computed tomography and magnetic resonance imaging detect cancers of internal organs.

B. Chemotherapeutics drugs are used to kill non-cancerous cells.

C. α-interferon activate the cancer patients' immune system and helps in destroying the tumour.

D. Chemotherapeutic drugs are biological response modifiers.

E. In the case of leukemia blood cell counts are decreases.

Choose the correct answer from the options given below: (NEET 2025)

(a) C and D only (b) A and C only

(c) B and D only (d) D and E only

b

After maturation, in primary lymphoid organs, the lymphocytes migrate for interaction with antigens to secondary lymphoid organ(s)/tissue(s) like:

A. thymus B. bone marrow
C. spleen D. lymph nodes
E. Peyer's patches

Choose the correct answer from the options given below: (NEET 2025)

(a) E, A, B only (b) C, D, E only
(c) B, C, D only (d) A, B, C only

b

Which of the following type of immunity is present at the time of birth and is a non-specific type of defence in the human body? (NEET 2025)

(a) Cell-mediated Immunity

(b) Humoral Immunity

(c) Acquired Immunity

(d) Innate Immunity

d

Identify the statement that is NOT correct.

(NEET 2025)

(a) Antigen binding site is located at C-terminal region of antibody molecules.

(b) Constant region of heavy and light chains are located at C-terminus of antibody molecules.

(c) Each antibody has two light and two heavy chains.

(d) The heavy and light chains are held together by disulfide bonds.

A

Match List I with List II: (NEET 2024)

List-I	List-II
A. Common cold	I. *Plasmodium*
B. Haemozoin	II. Typhoid
C. Widal test	III. Rhinoviruses
D. Allergy	IV. Dust mites

Choose the correct answer from the options given below:

(a) A-I, B-III, C-II, D-IV

(b) A-III, B-I, C-II, D-IV

(c) A-IV, B-II, C-III, D-I

(d) A-II, B-IV, C-III, D-I

Match List I with List II: (NEET 2024)

List-I	List-II
A. Cocaine	I. Effective sedative in surgery
B. Heroin	II. *Cannabis sativa*
C. Morphine	III. *Erythroxylum*
D. Marijuana	IV. *Papaver somniferum*

Choose the correct answer from the options given below:

(a) A-I, B-III, C-II, D-IV

(b) A-II, B-I, C-III, D-IV

(c) A-III, B-IV, C-I, D-II

(d) A-IV, B-III, C-I, D-II

C

Match List I with List II: (NEET 2024)

List-I	List-II
A. Typhoid	I. Fungus
B. Leishmaniasis	II. Nematode
C. Ringworm	III. Protozoa
D. Filariasis	IV. Bacteria

Choose the correct answer from the options given below:

(a) A-IV, B-III, C-I, D-II

(b) A-III, B-I, C-IV, D-II

(c) A-II, B-IV, C-III, D-I

(d) A-I, B-III, C-II, D-IV

Match List-I with List-II. (NEET 2023)

List-I		List-II	
A.	Ringworm	I.	*Haemophilus influenza*
B.	Filariasis	II.	*Trichophyton*
C.	Malaria	III.	*Wuchereria bancrofti*
D.	Pneumonia	IV.	*Plasmodium vivax*

Choose the correct answer from the options given below:

(a) A–II, B–III, C–I, D–IV

(b) A–III, B–II, C–I, D–IV

(c) A–III, B–II, C–IV, D–I

(d) A–II, B–III, C–IV, D–I

Match List-I with List-II. (NEET 2023)

List-I		List-II	
A.	Heroin	I.	Effect on cardiovascular system
B.	Marijuana	II.	Slows down body function
C.	Cocaine	III.	Painkiller
D.	Morphine	IV.	Interfere with transport of dopamine

Choose the correct answer from the options given below:

(a) A–I, B–II, C–III, D–IV

(b) A–IV, B–III, C–II, D–I

(c) A–III, B–IV, C–I, D–II

In which blood corpuscles, HIV undergoes replication and produces progeny viruses?

(NEET 2023)

(a) B-lymphocytes
(b) Basophils
(c) Eosinophils
(d) TH cells

d

Given below are two statements:

Statement I: Autoimmune disorder is a condition where the body defense mechanism recognizes its own cells as foreign bodies.

Statement II: Rheumatoid arthritis is a condition where the body does not attack self cells.

In the light of the above statements, choose the most appropriate answer from the options given below:

(NEET 2022)

(a) Both Statement I and Statement II are correct

(b) Both Statement I and Statement II are incorrect

(c) Statement I is correct but Statement II is incorrect

(d) Statement I is incorrect but Statement II is correct

C

Select the incorrect statement with respect to acquired immunity: (NEET 2022)

(a) Primary response is produced when our body encounters a pathogen for the first time.

(b) Anamnestic response is elicited on subsequent encounters with the same pathogen

(c) Anamnestic response is due to memory of first encounter

(d) Acquired immunity is non-specific type of defense present at the time of birth

d

Match List-I with List-II. (NEET 2021)

List-I		List-II	
(A)	Filariasis	(i)	*Haemophilus influenza*
(B)	Amoebiasis	(ii)	*Trichophyton*
(C)	Pneumonia	(iii)	*Wuchereria bancrofti*
(D)	Ringworm	(iv)	*Entamoeba histolytica*

Choose the correction answer from the options given below.

	(A)	(B)	(C)	(D)
(a)	(iv)	(i)	(iii)	(ii)
(b)	(iii)	(iv)	(i)	(ii)
(c)	(i)	(ii)	(iv)	(iii)
(d)	(ii)	(iii)	(i)	(iv)

Chronic auto immune disorder affecting neuro muscular junction leading to fatigue, weakening and paralysis of skeletal muscle is called as:

(a) Arthritis

(b) Muscular dystrophy

(c) Myasthenia gravis

(d) Gout

Identify the wrong statement with reference to immunity. (NEET 2021)

(a) Active immunity is quick and gives full response ×

(b) Foetus receives some antibodies from mother, it is an example for passive immunity

(c) When exposed to antigen (living or dead) antibodies are produced in the host's body. It is called "Active immunity"

(d) When readymade antibodies are directly given, it is called "Passive immunity"

a

Match the following diseases with the causative organism and select the correct option. (NEET 2021)

Column I	Column II
(a) Typhoid	(i) *Wuchereria*
(b) Pneumonia	(ii) *Plasmodium*
(c) Filariasis	(iii) *Salmonella*
(d) Malaria	(iv) *Haemophilus*

(a) a – ii; b – i; c – iii; d – iv

(b) a – iv; b – i; c – ii; d – iii

(c) a – i; b – iii; c – ii; d – iv

(d) a – iii; b – iv; c – i; d – ii

Match the following columns and select the correct option. (NEET2020)

Column I	Column II
(a) Eosinophils	(i) Immune response
(b) Basophils	(ii) Phagocytosis
(c) Neutrophils	(iii) Release histaminases, destructive enzymes
(d) Lymphocytes	(iv) Release granules containing histamine

(a) a – i; b – ii; c – iv; d – iii

(b) a – ii; b – i; c – iii; d – iv

(c) a – iii; b – iv; c – ii; d – i

(d) a – iv; b – i; c – ii; d – iii

(c)

The infectious stage of *Plasmodium* that enters the human body is [2020]

(a) Female gametocytes (b) Male gametocytes

(c) Trophozoites (d) Sporozoites

d

Identify the correct pair representing the causative agent of typhoid fever and the confirmatory test for typhoid. (NEET2019)

(a) *Streptococcus pneumoniae*/Widal test

(b) *Salmonella typhi*/Anthrone test

(c) *Salmonella typhi*/Widal test

(d) *Plasmodium vivax*/UTI test

c

Which of the following immune responses is responsible for rejection of kidney graft? [2019]

(a) Humoral immune response

(b) Inflammatory immune response

(c) Cell-mediated immune response

(d) Auto-immune response

c

Drug called 'Heroin' is synthesized by **(NEET2019)**

(a) acetylation of morphine

(b) glycosylation of morphine

(c) nitration of morphine

(d) methylation of morphine

a

Which part of poppy plant is used to obtain the drug "Smack"? (NEET2018)

(a) Roots
(b) Latex
(c) Flowers
(d) Leaves

In which disease does mosquito transmitted pathogen cause chronic inflammation of lymphatic vessels? (NEET2018)

(a) Ringworm disease
(b) Ascariasis
(c) Elephantiasis
(d) Amoebiasis

Elephantiasis is also called lymphatic filariasis. It is caused by filarial worms. They build up in lymphatic vessels blocking their flow and causing inflammation.

Which of the following is not an autoimmune disease? (NEET 2018)

(a) Alzheimer's disease
(b) Rheumatoid arthritis
(c) Psoriasis
(d) Vitiligo

Evolution

New NCERT

Origin of Universe = $\boxed{13.8}$ B.Y.A (According to Big Bang theory).

Origin of Earth = 4.5 B.Y.A

Origin of Life = 4.0 B.Y.A [500 million after formation of Earth]

- Non cellular ((3.0 B.Y.A)
  - Giant Molecules
    - R.N.A → DNA
    - Protein
    - Polysaccharids.
    - etc
- Cellular [2.0 B.Y.A]
  - Unicellular
    ↓
  - Multicellular

Theories to Explain Origin of Life

1. Theory of Special Creation.
2. Theory of Catastrophism.
3. Theory of Panspermia/Cosmozoic theory - Some scientists believe that it came from outside. Early Greek thinkers thought units of life called spores were transferred to different planets including earth. 'Panspermia' is still a favourite idea for some astronomers.
4. Theory of Spontaneous Generation (Theory of abiogenesis)- (Aristotle and Von Helmont)

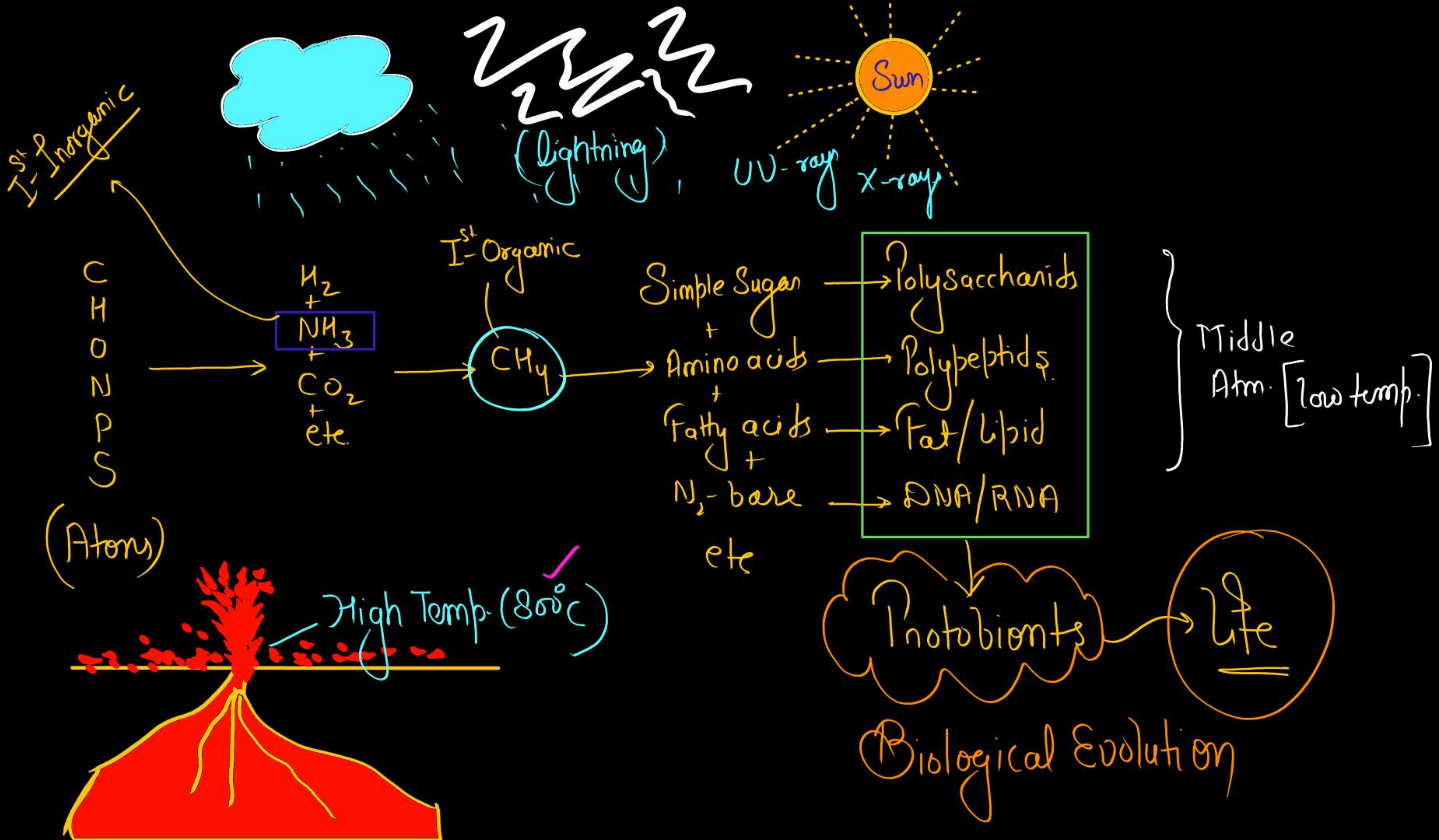
- According to this life arises spontaneously out of non living matter.
- Von Helmont - Within 21 days, rats are produced from wheat grains kept with a shirt drenched in sweat.

5. Theory of Biogenesis.

- The theory of spontaneous generation was disproved by many scientists.
- They proved that new organisms can be formed from pre-existing ones, i.e., omnis vivum ex vivo. Noted scientists who experimentally challenged the theory were Francesco Redi, Spallanzani and Louis Pasteur.

6. Oparin-Haldane Theory. vvvImp

or [Pri. Abiogenesis or (Chemical Evolution)]



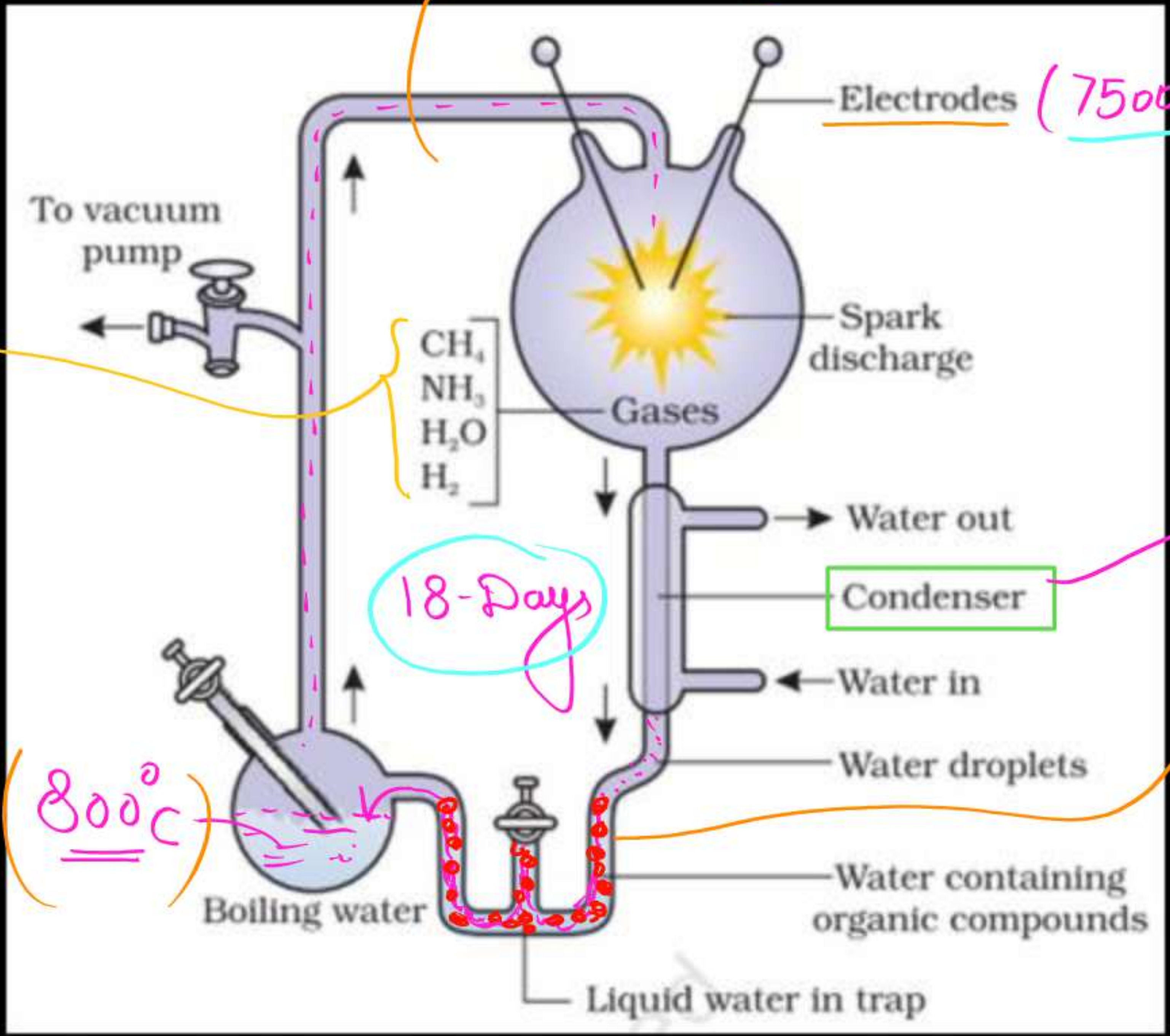
Ist Inorganic
(lightning)
Sun
UV-ray
X-ray
Ist Organic
C
H
O
N
P
S
(Atoms)
H_2 + NH_3 + CO_2 + etc.
CH_4
Simple Sugar → Polysaccharids
+ Amino acids → Polypeptids
+ Fatty acids → Fat/Lipid
+ N_2-base → DNA/RNA
etc
Middle Atm. [low temp.]
High Temp. (800°C)
Protobionts → Life
Biological Evolution

Miller's Experiment- $\boxed{(1953)}$ Q

Devoid of Radiation (UV/X-Rays)

Gases

Q// $\boxed{\begin{matrix} CH_4 : NH_3 : H_2 \\ 2 : 1 : 2 \end{matrix}}$



(75000 Volt)

18-Days

Q/ (800°C)

Q// Low Temp. (Middle Atm)

Q// [Amino Acids]

Glycine ✓
\+
Alanine ✓
\+
Aspartic acid. ✓

Origin of Prokaryotes

3.0 BYA [Non cellular Life] — Formation of Protobionts [Protein, Carbohydrate, RNA etc]

↓

2.0 BYA [Cellular Life] — Prokaryotes

1) - Chemoheterotroph — Fermentation of Organic Sub. → Energy

2) - Chemoautotroph (Iron Bacteria) — Chemical Reaction (O/R) → Energy

3) - Simple photoautotroph. [Sulphar Bacteria (Anaerobic Res.)]
- Development of Pigments
- → use - light Energy
- → Source of $H^+ = H_2S$

4) - Cynobacteria ⇒ Source of $H^+ = H_2O$

$2H_2O \longrightarrow O_2 + 4H^+$

(Release O_2)

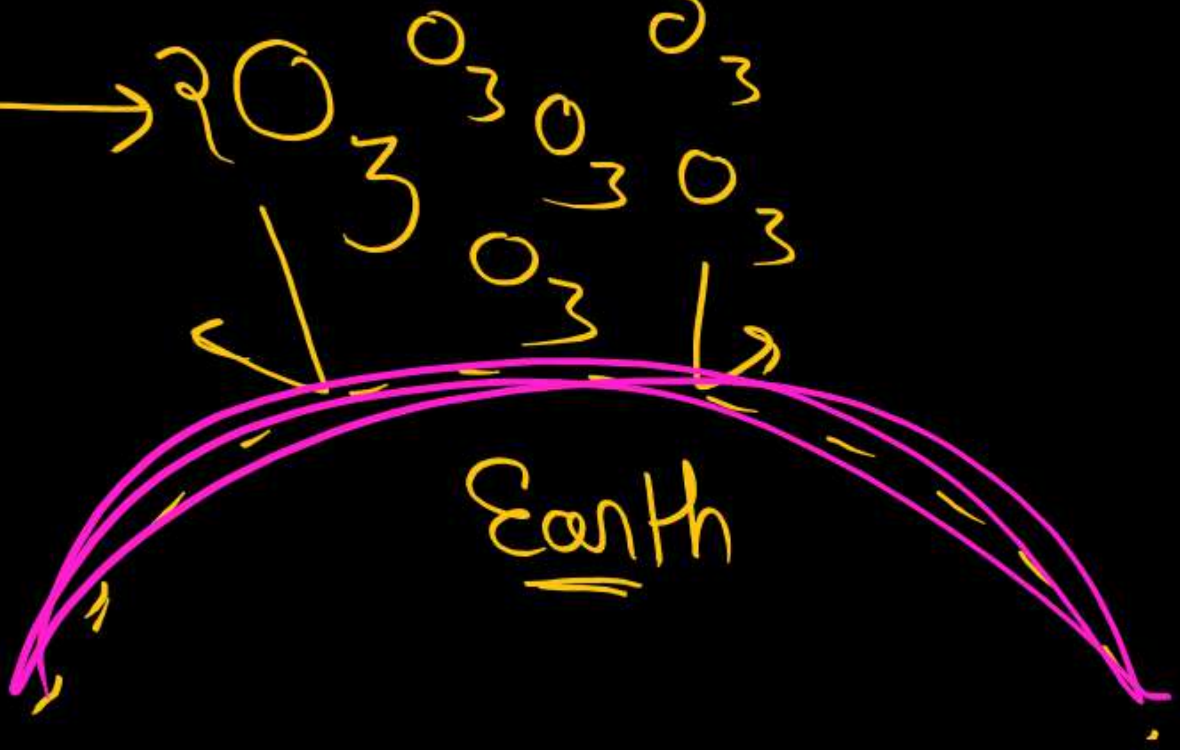
$$2H_2O \longrightarrow \underset{\substack{\downarrow \\ ATP}}{4H^+} + O_2 \uparrow$$

[oxydizing Atm.]

$$2O_2 + \boxed{CH_4} \longrightarrow \boxed{CO_2} + 2H_2O$$

$$3O_2 + \boxed{4NH_3} \longrightarrow \boxed{2N_2} + 6H_2O$$

$$2O_2 + O_2 \longrightarrow 2O_3$$



(1) Palaeontological Evidences (Evidences from Fossil Record)

Fossils - Fossils can be defined as the remains or traces of the non-degraded parts of ancient living things found within rock life preserved in the Earth's crust.

→ Sedimentory rocks

Age of Fossils

Radioactive dating Tech. →

R.A. (1000 gm) → R.A (500) → RA (250), NRA (250)

R.A. (1000 gm) → NRA (500)

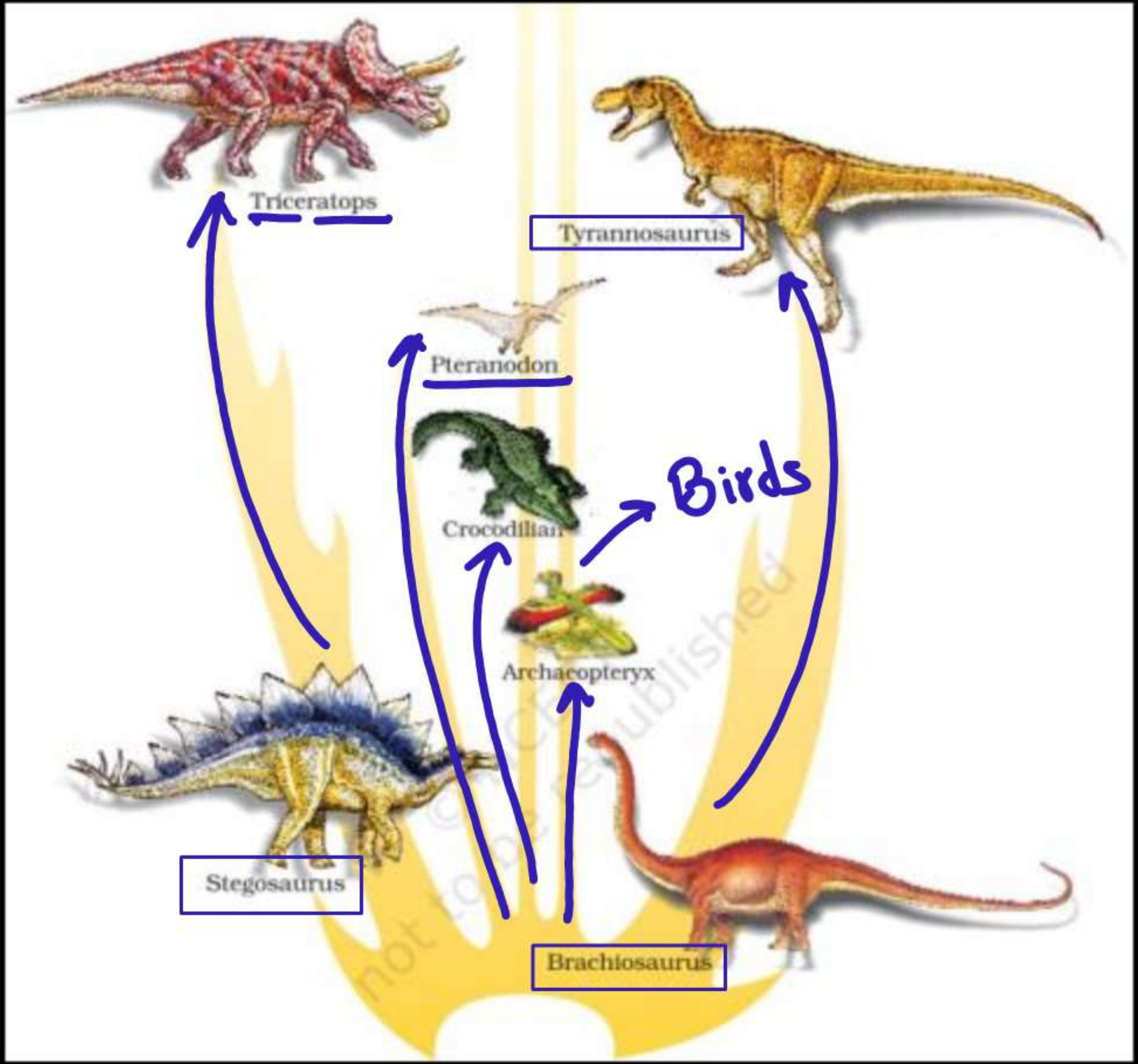
1 $\frac{1}{2}$ life

2nd $\frac{1}{2}$ life

Ratio of RA : NRA in —

1 $\frac{1}{2}$ life = 1 : 1

2 $\frac{1}{2}$ life = 1 : 3



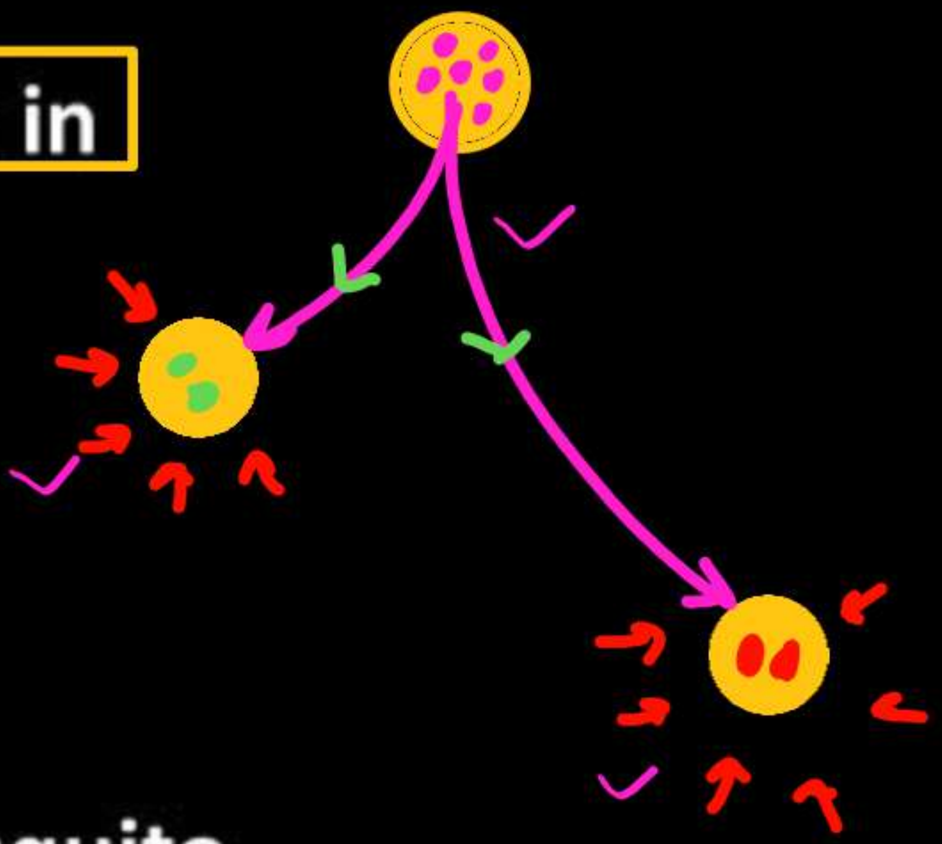
Triceratops
Tyrannosaurus
Pteranodon
Crocodilian
Birds
Archaeopteryx
Stegosaurus
Brachiosaurus

(2) Evidences from Comparative Anatomy and Morphology

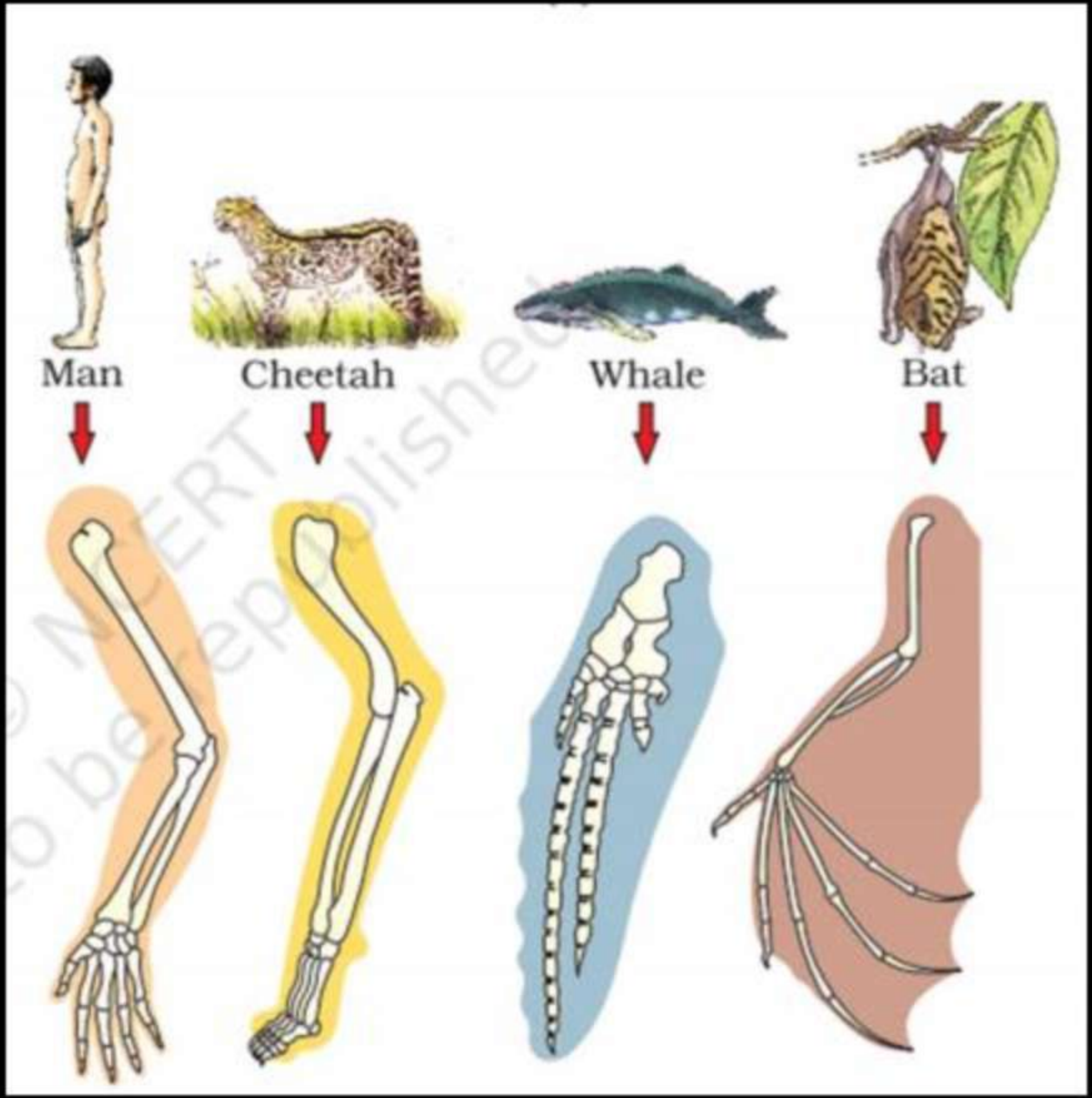
1 Q-100%.

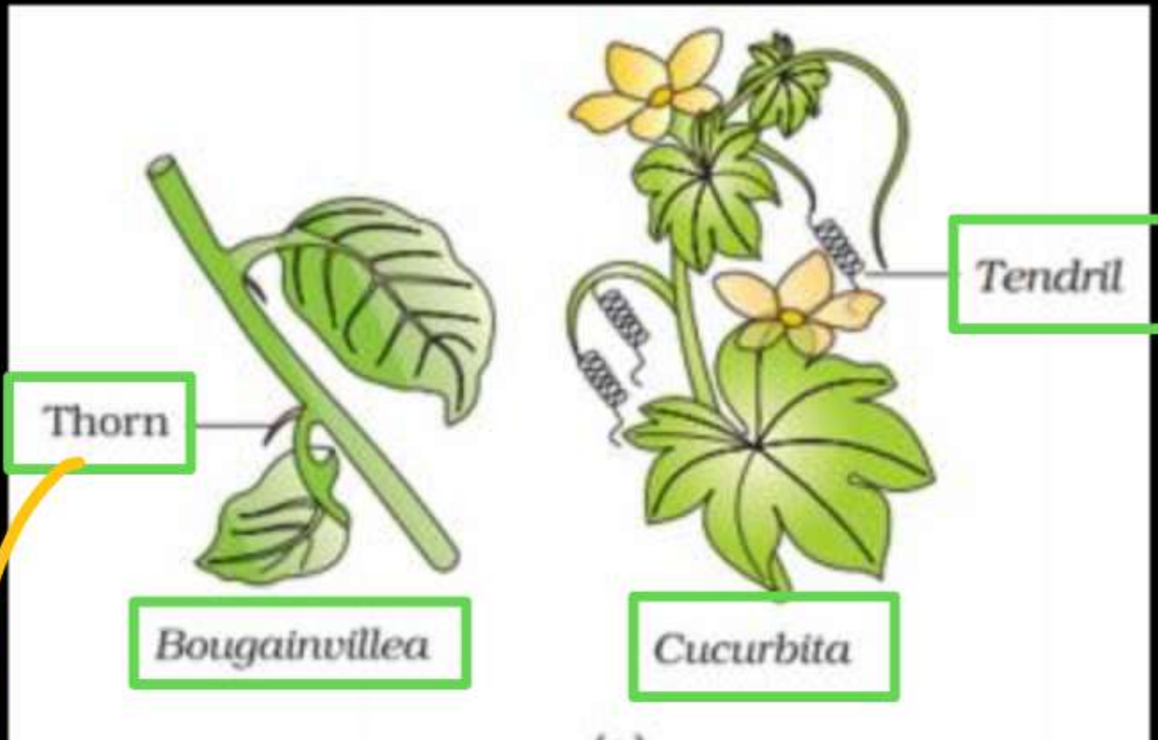
Homologous organs-

- Similar fundamental structure but are different in functions
- Divergent evolution
- Common ancestry
- Eg.-



1. Forelimbs of man, cheetah, whale and bat.
2. The mouth parts of cockroach, honey bee, mosquito and butterfly.
3. Thorn and tendrils of *Bougainvillea* and *Cucurbita* represent homology





(a)

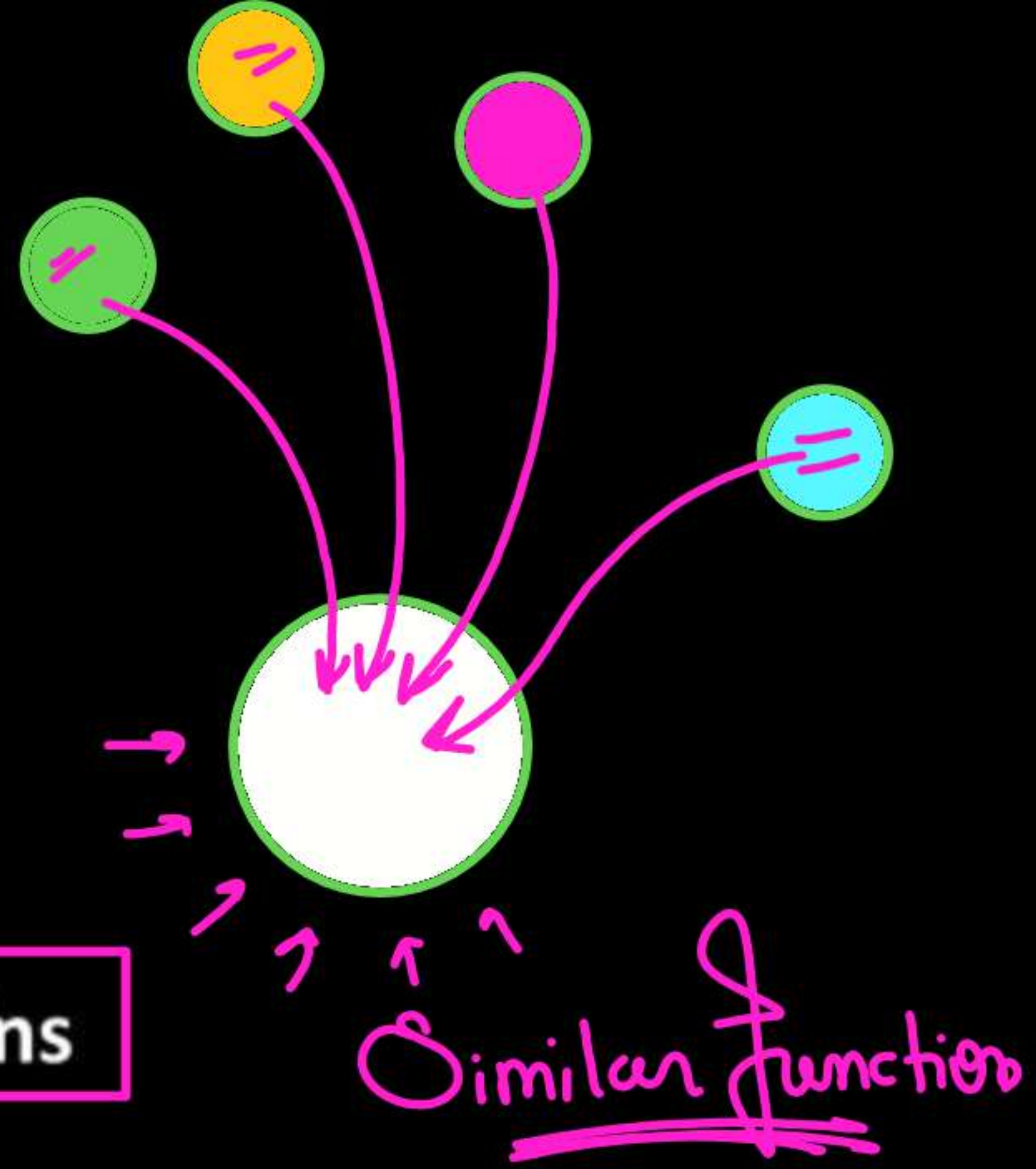
Protection

Support

Analogous Organs

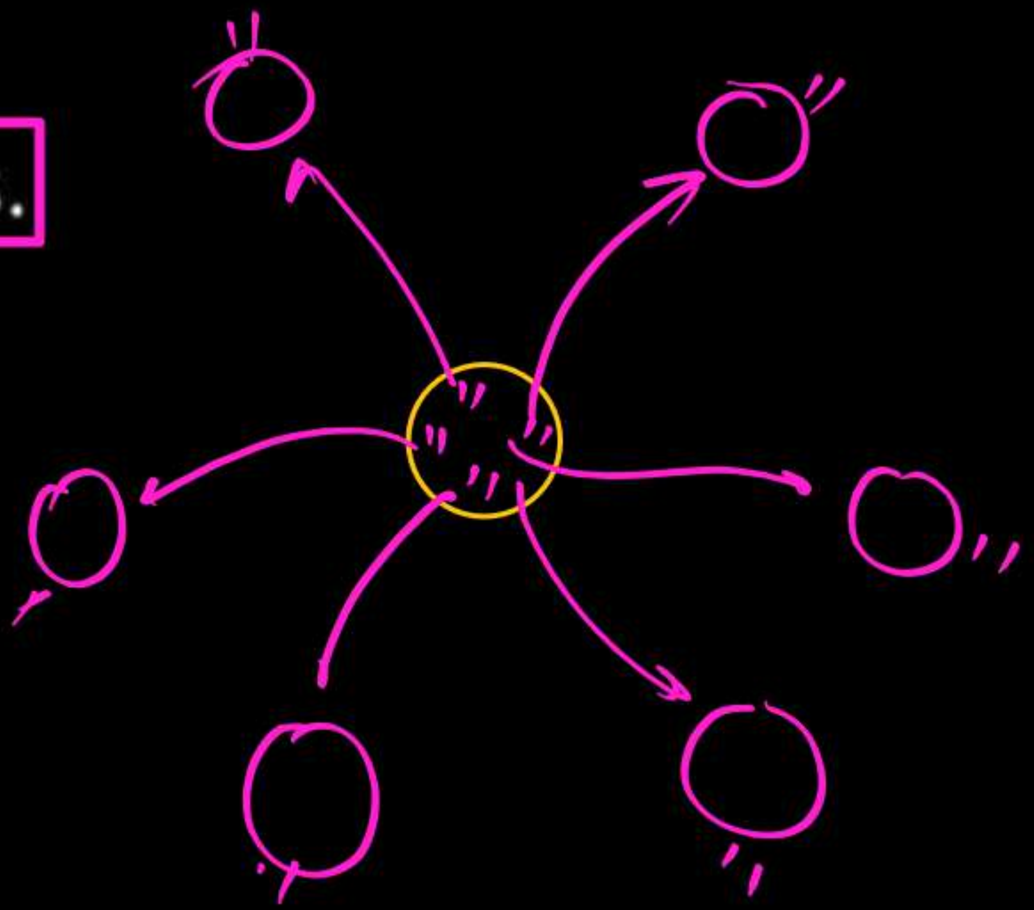
- Different their anatomical structural but
- Similar in functions due to same needs.
- Result of convergent evolution
- Eg.-

1. Wings of butterfly and of birds
2. Pectoral Fins of sharks and flippers of Dolphins
3. Stings of honey bee and scorpion
4. Eye of octopus and eye of mammals
5. Sweet potato (root modification) and potato (stem modification)



ADAPTIVE RADIATION –

- During his journey (1831) Darwin went to Galapagos Islands - 22.
- Ship - “H.M.S. Beagle
- Duration of journey 5 years.



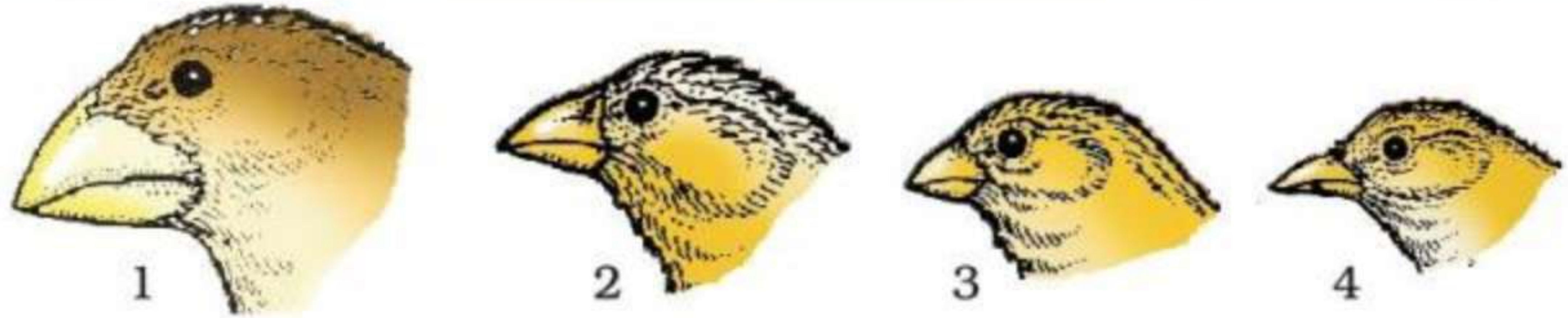
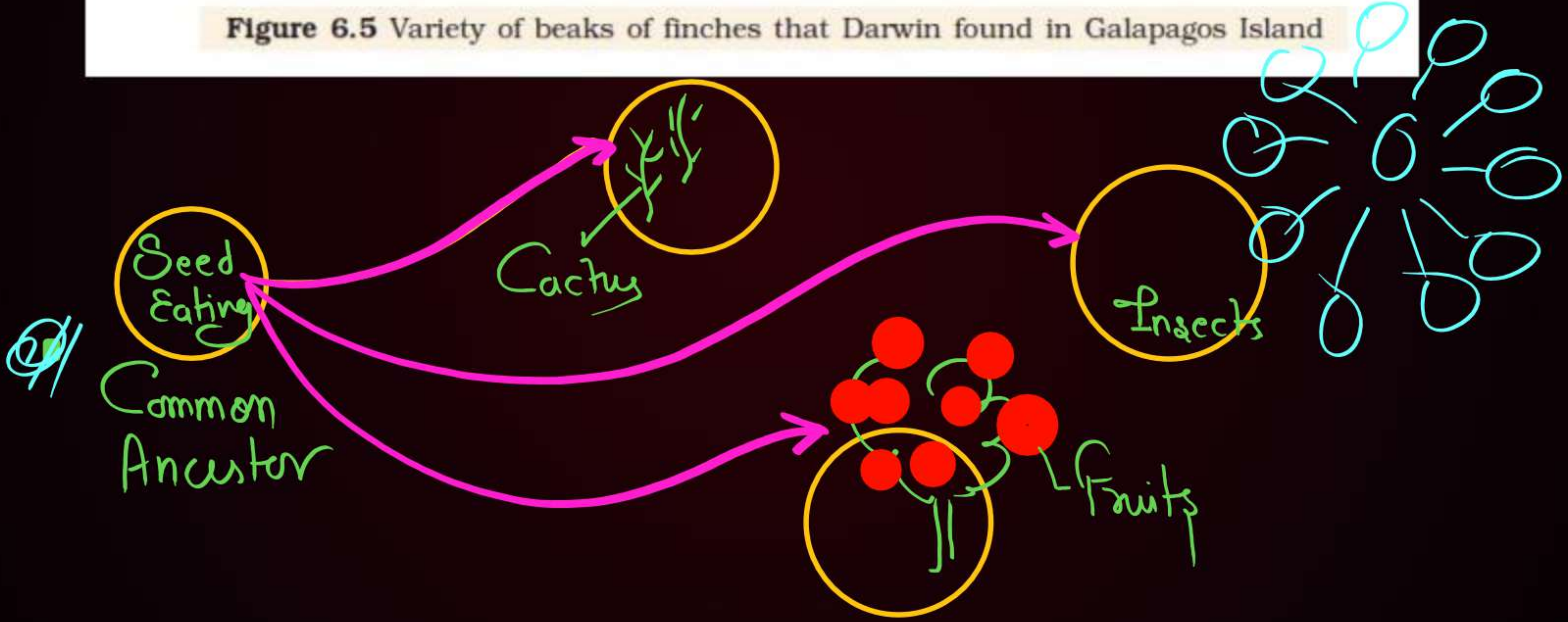


Figure 6.5 Variety of beaks of finches that Darwin found in Galapagos Island



Australian marsupials -

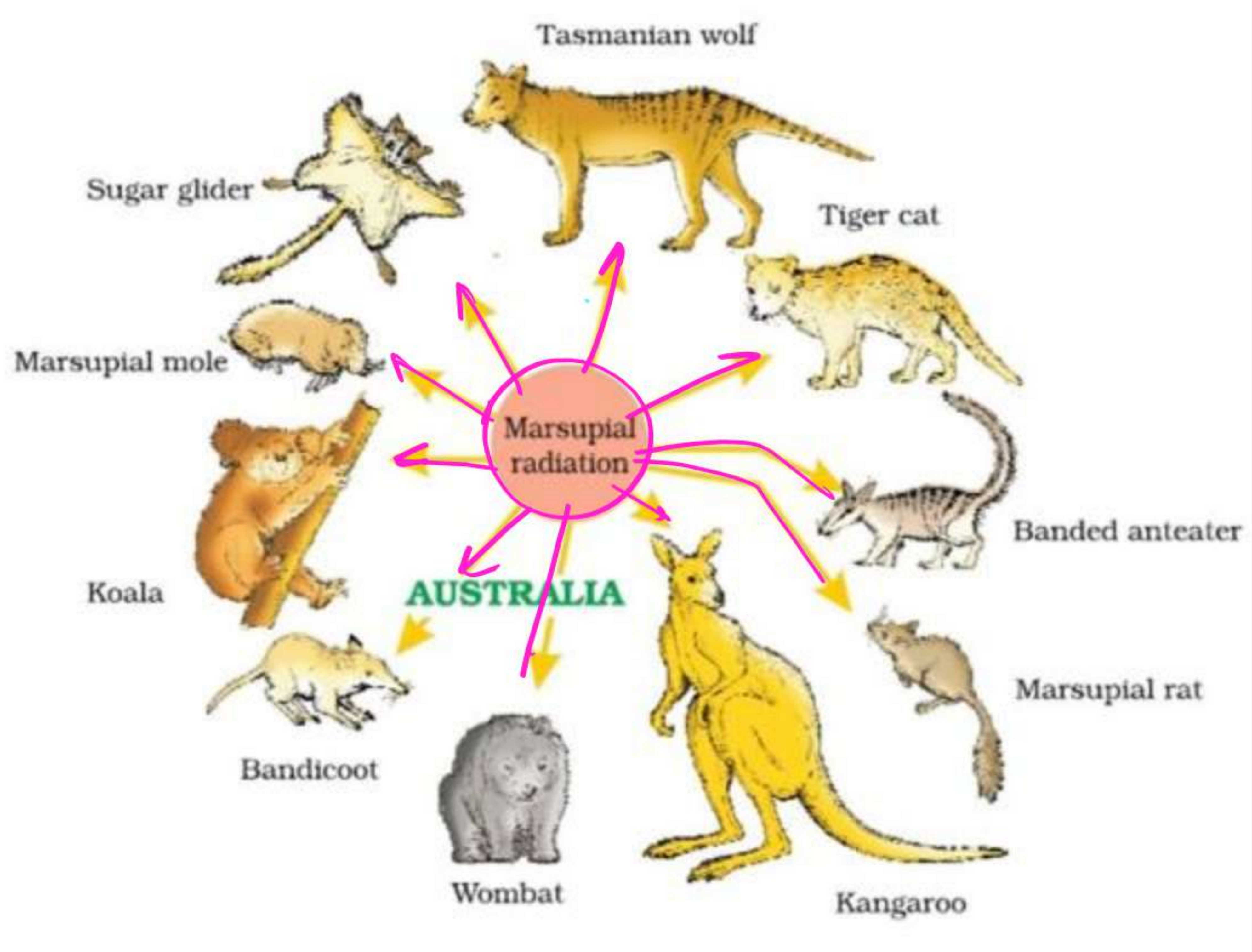
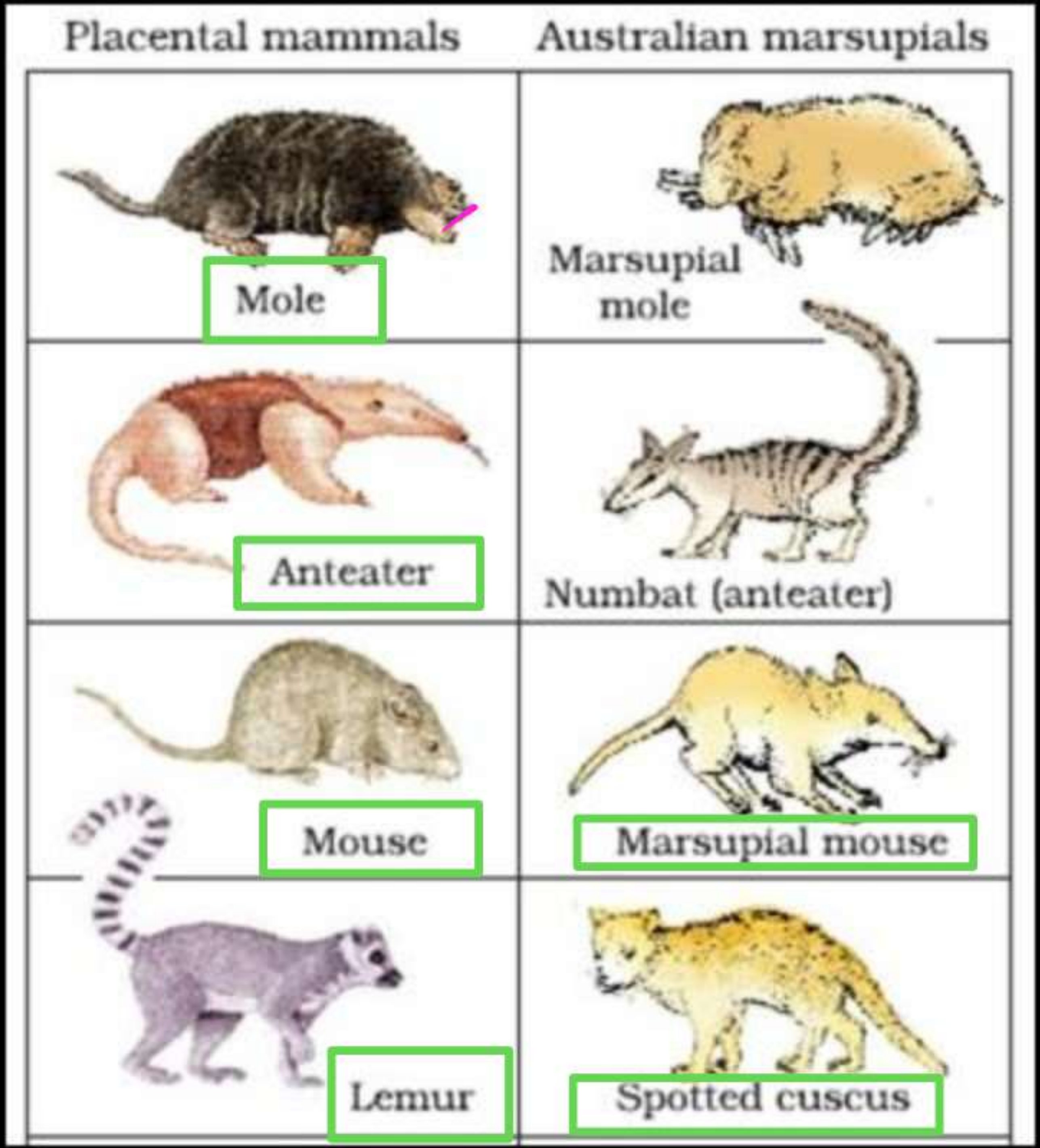


Figure 7.6 Adaptive radiation of marsupials of Australia



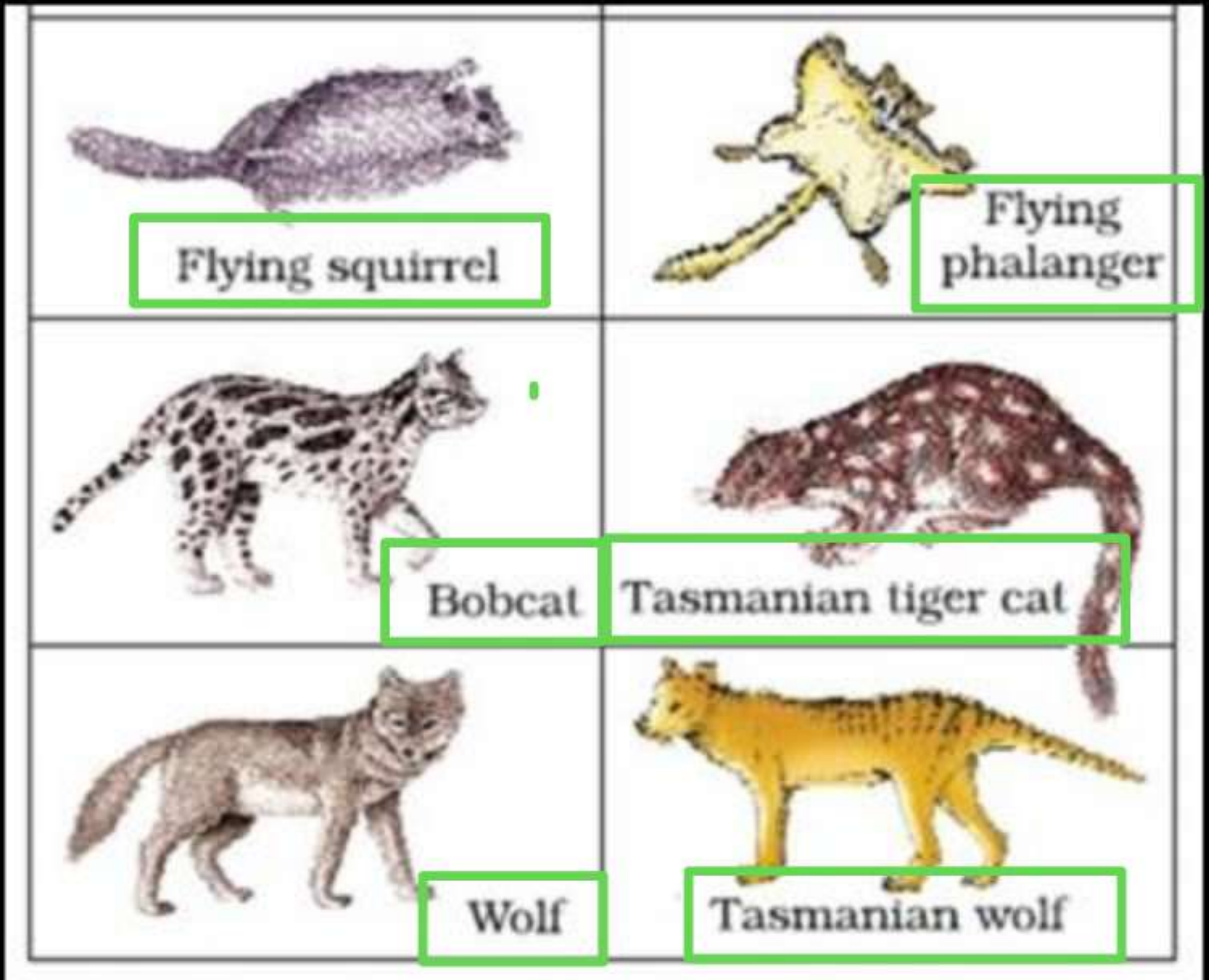
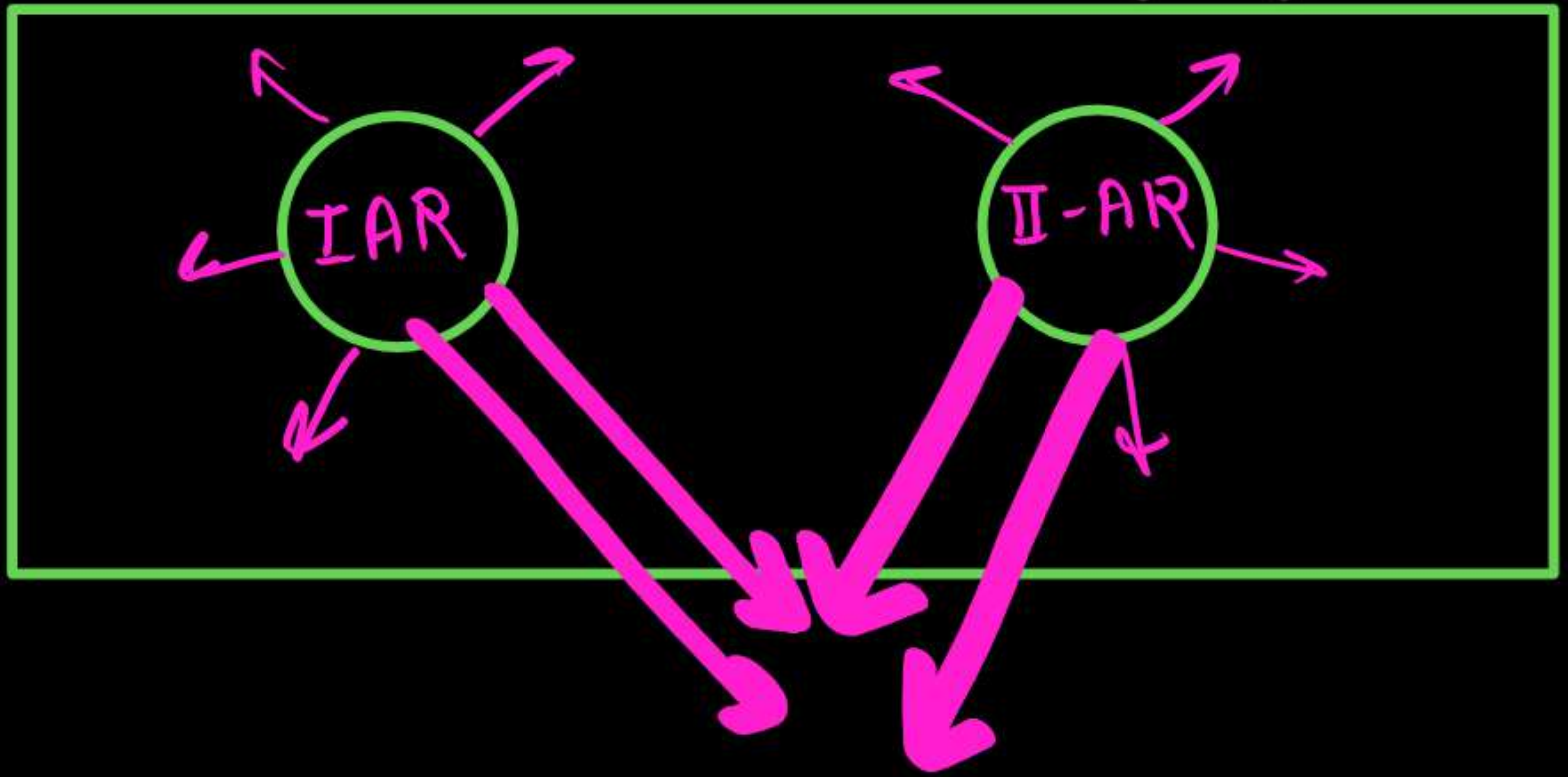


Figure 7.7 Picture showing convergent evolution of Australian Marsupials and placental mammals

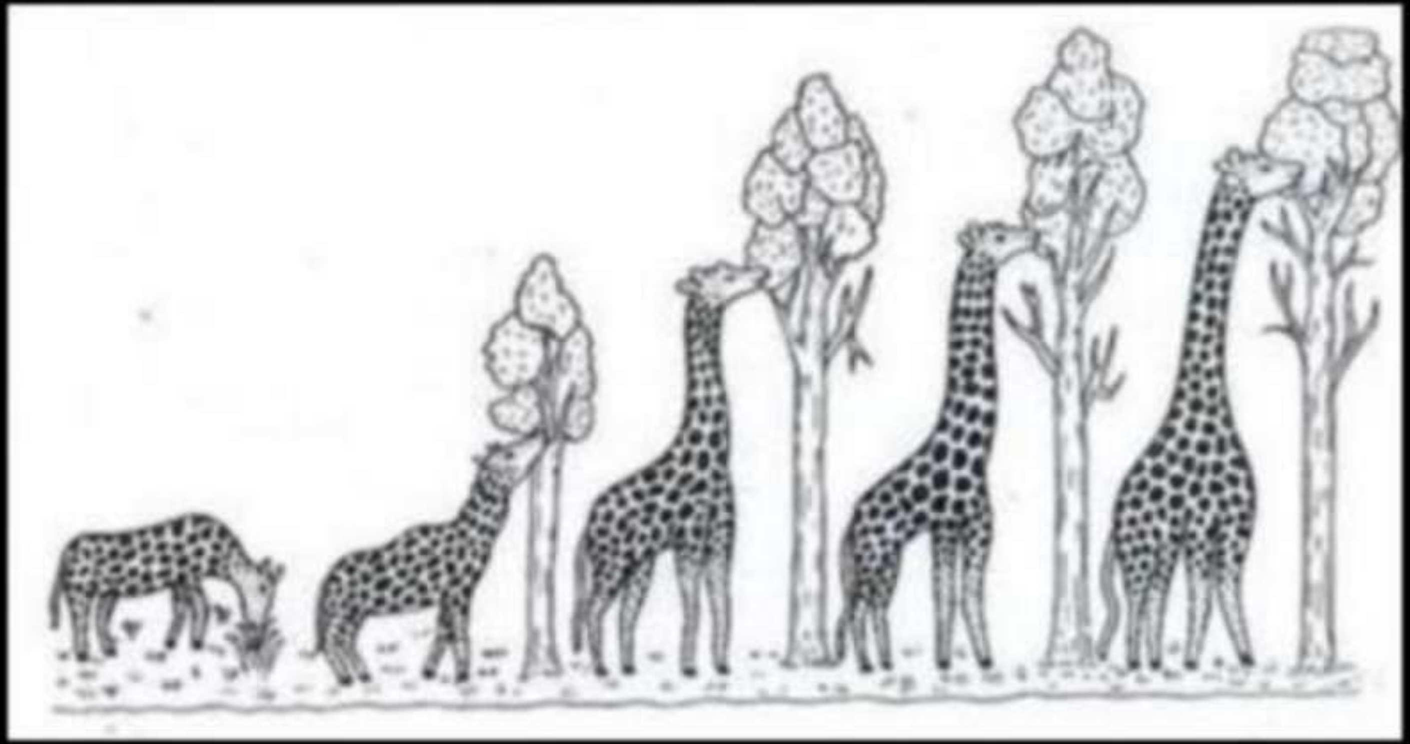
- When more than one adaptive radiation appeared to have occurred in an isolated geographical area (representing different habitats), one can call this convergent evolution.
- Placental mammals in Australia also exhibit adaptive radiation in evolving into varieties of such placental mammals each of which appears to be 'similar' to a corresponding marsupial (e.g., Placental wolf and Tasmanian wolf-marsupial).



Biological Evolution –

(1) Lamarck's Theory of Evolution -

* Theory of use & disuse of Organs
or
× Inheritance of Acquired character



(2) Darwinism -

Imp

- Theory of Natural Selection or Origin of Species } (1859)

- Darwin was influenced by ⇒

  An Essay on principle of Population = By = T.R. Malthus

- Alfred Walloce = Naturalist, = Malay Archipelago

  [Similar Conclusion around same times]

Main points of Theory of Natural Selection

1) - High rate of Reproduction ✓

2) - Total number almost Constant ✓

3) - Struggle for existence [Competition]

[Intra specific]
(b/w member of Same Species)

[Interspecific]
[b/w Member of different Species]

4) - Variation & Heridity → [Continuous Variation] θ

[Adaptive] ✓
(useful)

[Nonadaptive] ×
(Harmful)

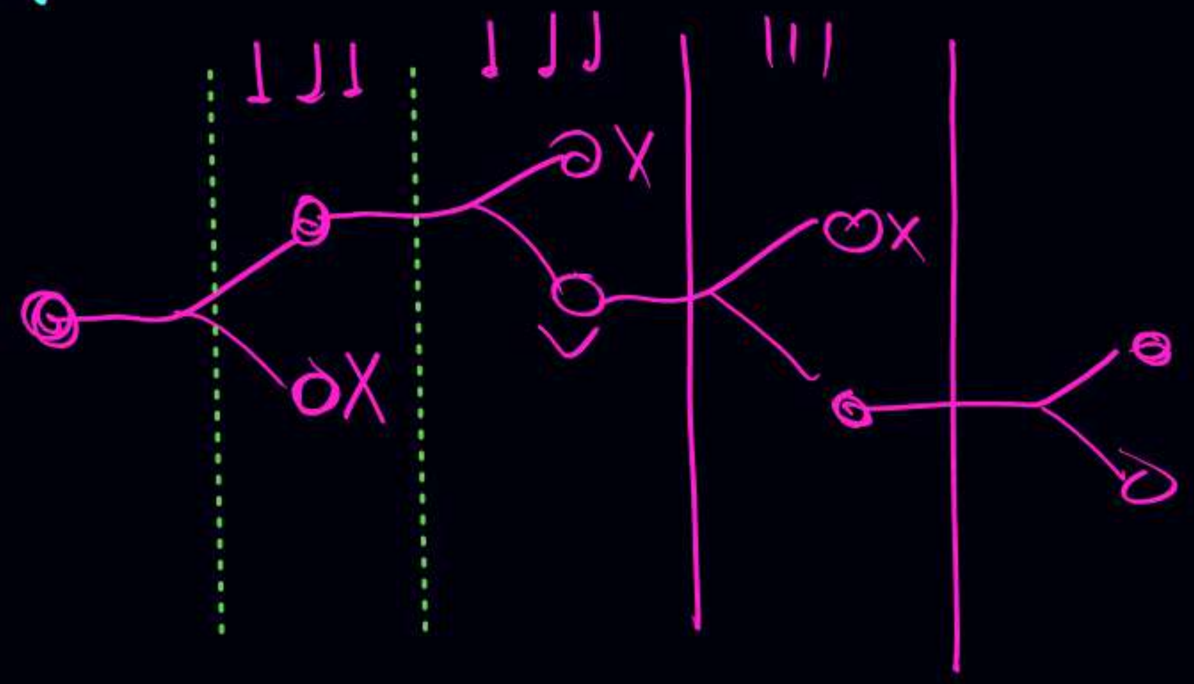
5) - Natural Selection → Individual with $\boxed{max^m \text{ useful Variation}}$ Selected by = Nature

6) - Survival of fittest → Fitness is $\boxed{\text{end result}}$ of the ability to adapt & get Selected by Nature.

* Two Key Concepts of Darwin theory

① [Branching Descent]
(Pattern of Evolution)

② [Natural Selection]
Mechanism of Evolution



Industrial Melanism

Peppered Moth
or
Biston butularia

[England]

[Before Industilisation]
[1850]

- ↑ No. of white wings Moth
- ↓ No. " Black " "

[After Industilization]
[1920]

- ↑ No. of Black wings Moth
- ↓ No. of White " "

Artificial Selection (Darwin)

Man has bred selected plants and animals for agriculture, horticulture, sport or security. Man has domesticated many wild animals and crops. This intensive breeding programme has created breeds that differ from other breeds (e.g., dogs) but still are of the same group. It is argued that if within hundreds of years, man could create new breeds, could not nature have done the same over millions of years?

(3) Mutation Theory → Hugo de Vries

↳ work on - Evening primrose
or (Oenothera lamarckiana)

According to this theory →

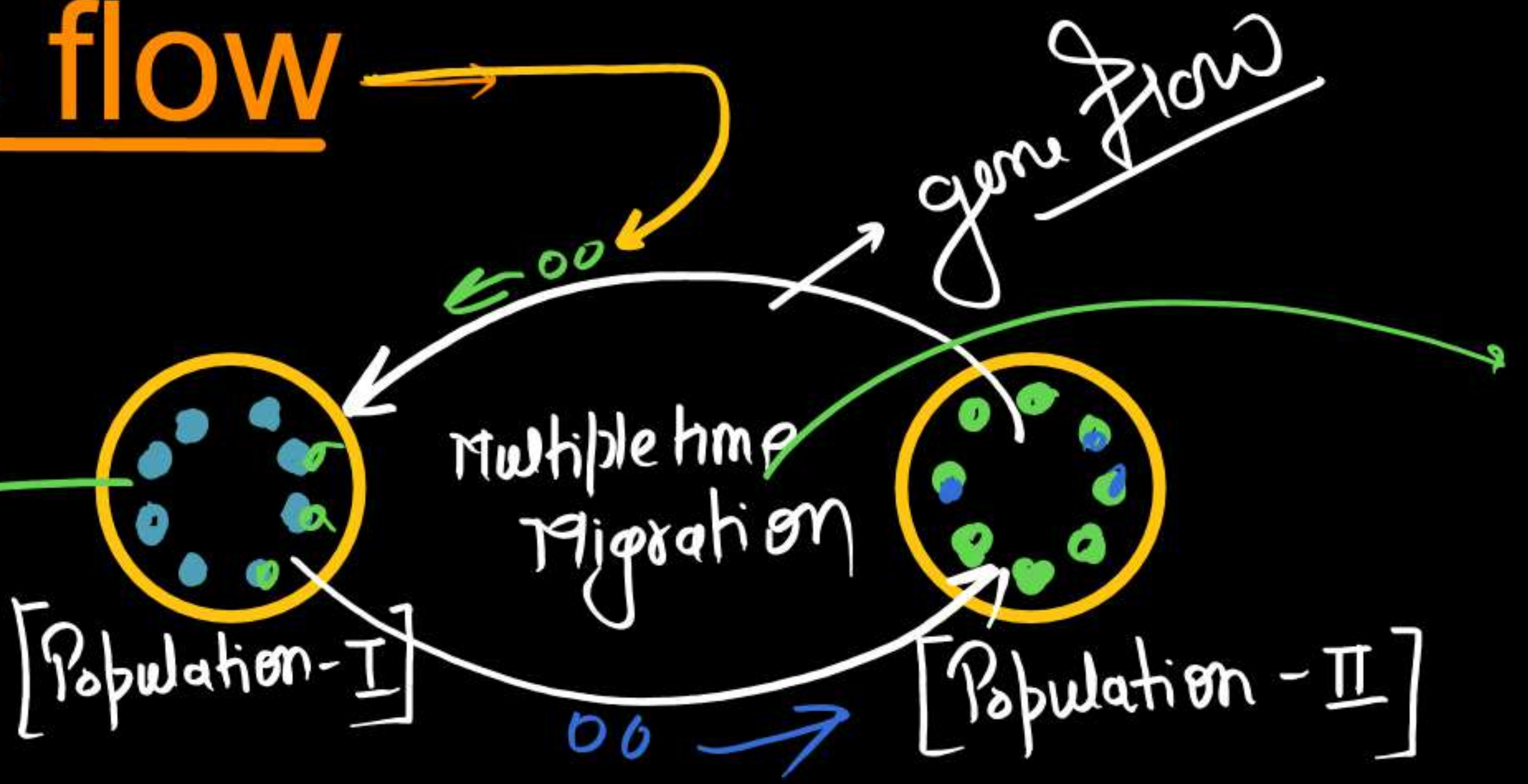
* Mutation are responsible for Evolution
* Mutation provide raw material for evolution
* Variation are large, Discontinuous & Nondirectional, Randum
* Evolution is Sudden & jerky Process. →
Called - Saltation (Single Step Large Mutation)

MODERN/SYNTHETIC THEORY

- Many factors are collectively responsible for formation of new species :⟶

1) - Gene migration/gene flow
2) - Genetic Drift
3) - Mutation
4) - Genetic Recombination
5) - Natural Selection

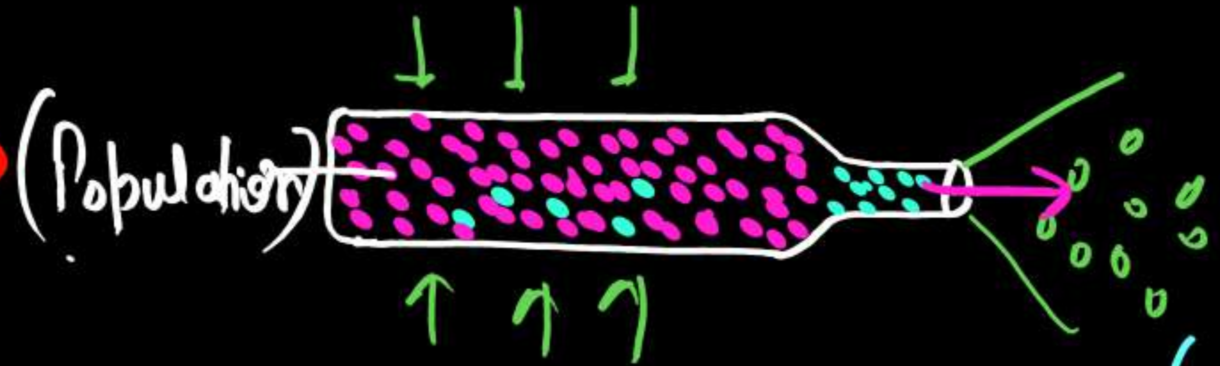
(1) Gene flow



Formation of New Species.

Gene pool → [Collection of all gene and its alleles in a population]

(2) Genetic Drift → Bottle Neck Effect



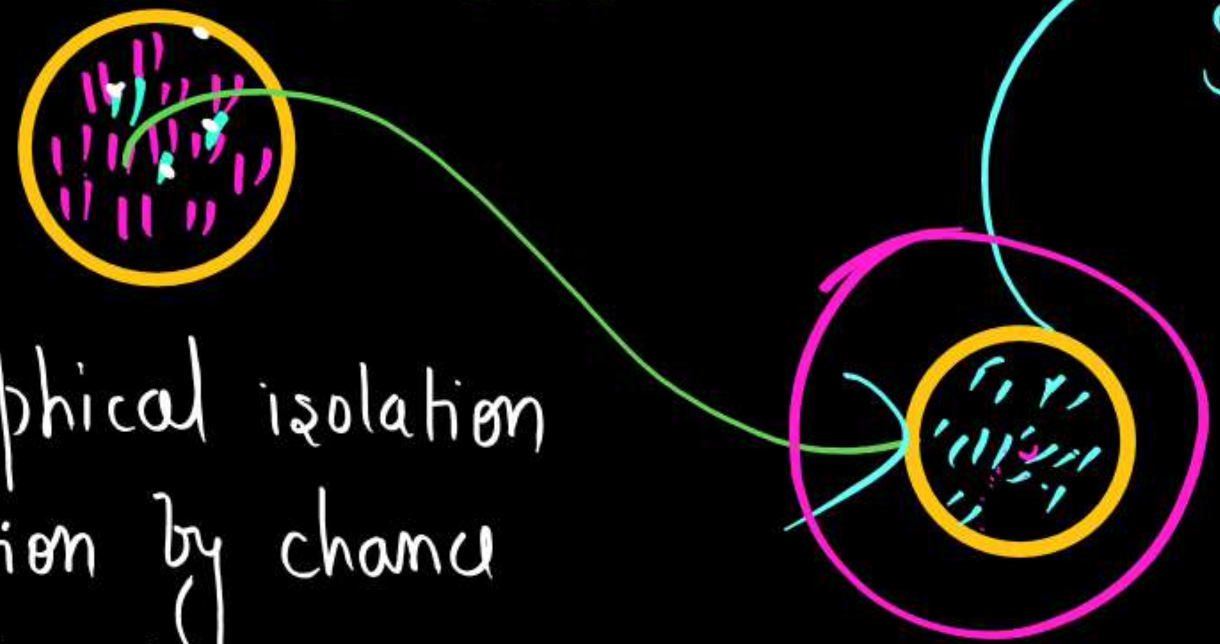
V imp

Small Population

⊙ Randum / Nondirectional changes of gene frequencies in small Population by Chance.

⊙ May - ↑ or ↓ Due to (Size of Population)

Founder effect



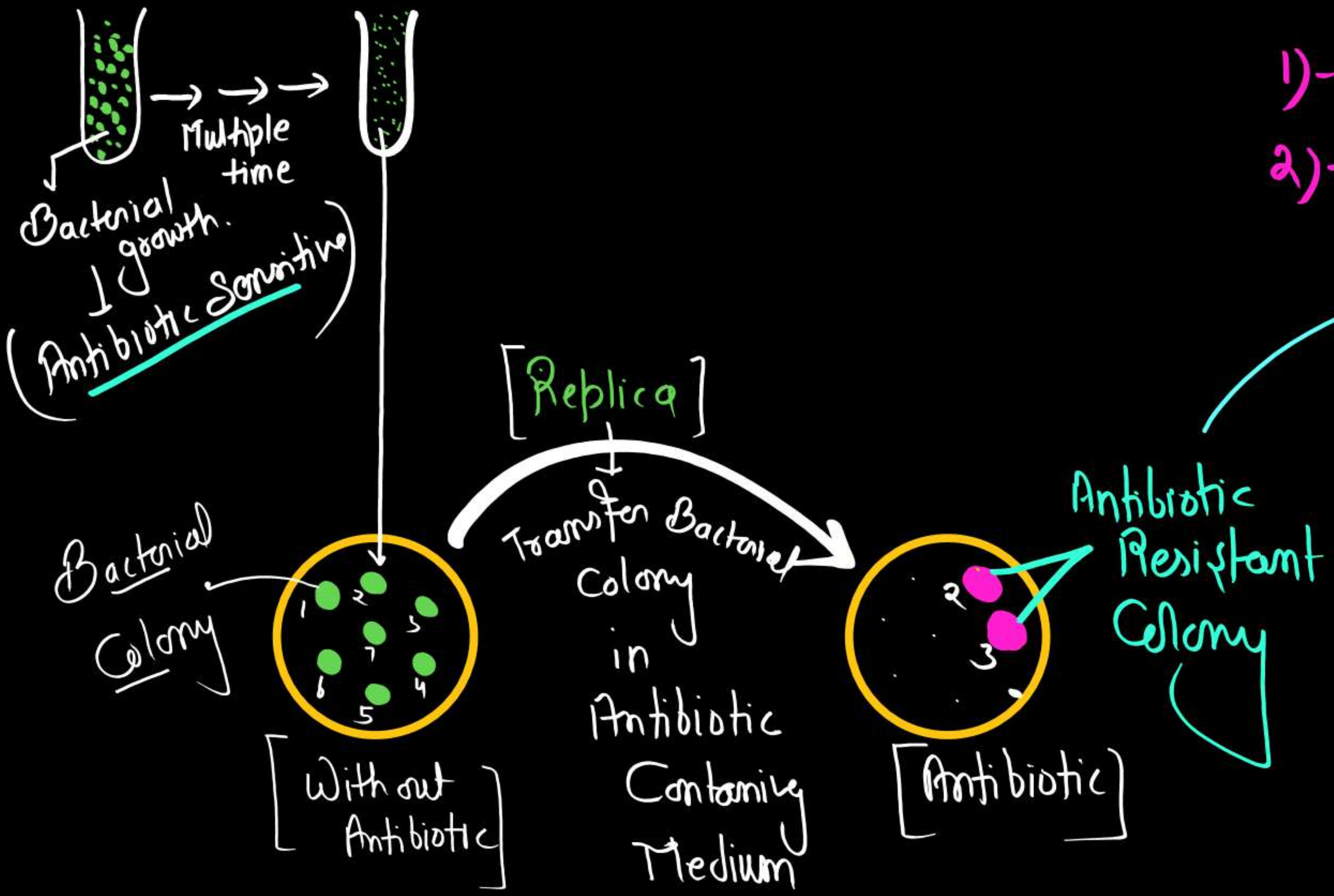
New Species.

- Geographical isolation
- Migration by chance
- Natural Calamity.

(3) Mutation/Genetic Basis of Adaptation

Lederberg's experiments →

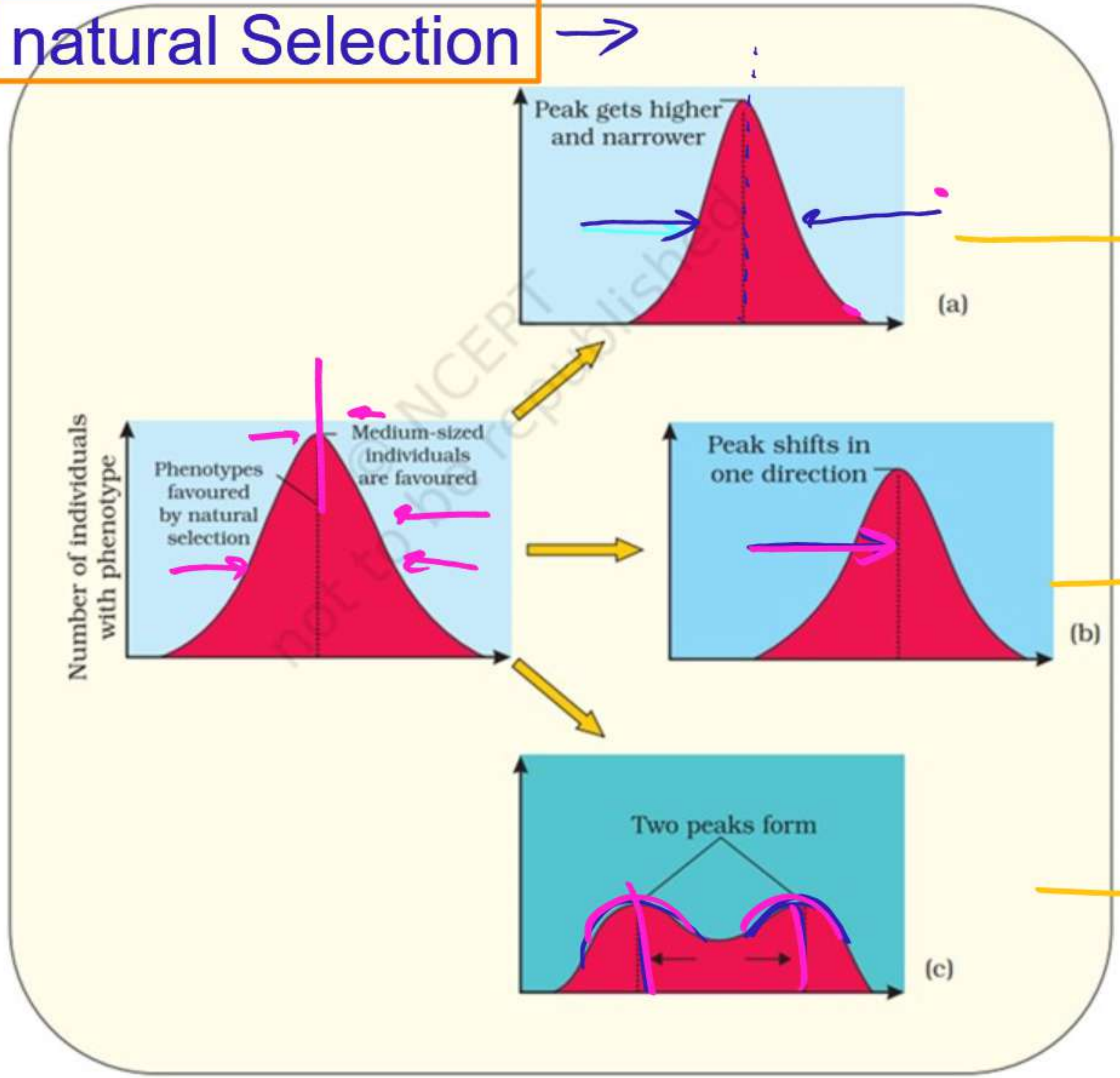
- Microbial experiments show that –
  Pre-existing advantageous mutation, when selected will result in observation of new phenotypes
- Over few generation, this would result in Speciation



1)- Antibiotic Sensitive

2)- Antibiotic Res.

natural Selection →



7.8 Diagrammatic representation of the operation of natural selection on different traits : (a) Stabilising (b) Directional and (c) Disruptive

Types of Natural Selection

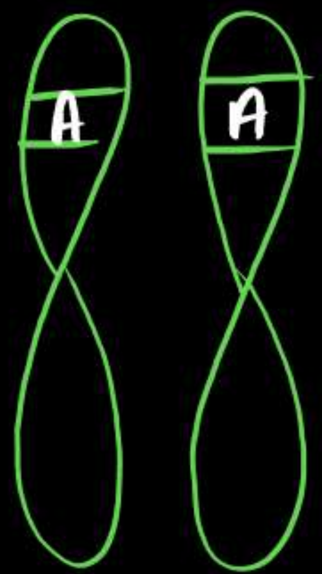
Stabilising Selection

Directional Selection

Disruptive Selection

HARDY-WEINBERG PRINCIPLE -

Population (Large) → Frequency of occurrence of alleles of gene or Locus or Individual



gene
- AA
- aa
- Aa

A = P

a = q

Gene and allilic frequency in a $\boxed{\text{large population}}$ remains Constant if —
- Randum Mating
- Absence of 5-Factors

1) - Mutation 2) Gene migration, 3) Recombination
4) - Genetic Drift 5) Natural Selection

$$p^2 + q^2 + 2pq = 1$$

or

$$(p+q)^2 = 1$$

or

$$(p+q=1)$$

A Brief account of Evolution →

- I^{st} acellular life = 3 BYA
- I^{st} cellular life = 2 BYA
- Invertebrate become active = 500 MYA
- I^{st} Vertebrate (jaw less fish) = 350 M.Y.A
- I^{st} jawed Vertebrate = 350 M.Y.A.

eg = Latemaria / Coelacanth / Lobe fin fish → Living fossil

→ Cough in South africa (1938)

→ Though to be extinct

I^{st} amphibia

1^{st} amphibia → (Extinct) = Invaded land after plant

(Frogs & Salamander)

Early Reptiles → Extinct

→ Dinosours = Extinct

→ Turtles
→ Lizards
→ Snakes
→ Tuatara
→ Crocodile

∴ Survived

Birds

Mammals

→ Some back in water = 200 mya
= Icthyosaurs.

* Dominating land reptile
in jurassic period = 150 mya

* Biggest Dinosaur → Tyrannosaurus rex (Hight = 20 feet

* About = 65 mya → Dinosaurs suddenly Disappeared from Earth [In Cretaceous period]

Modern Birds

Mammals

Flowering Plants [Angiosperms]

Period

4- Era.

Quaternary

Tertiary

Cretnceary

Jurasic

Triasic

Permean

Carbonifenous

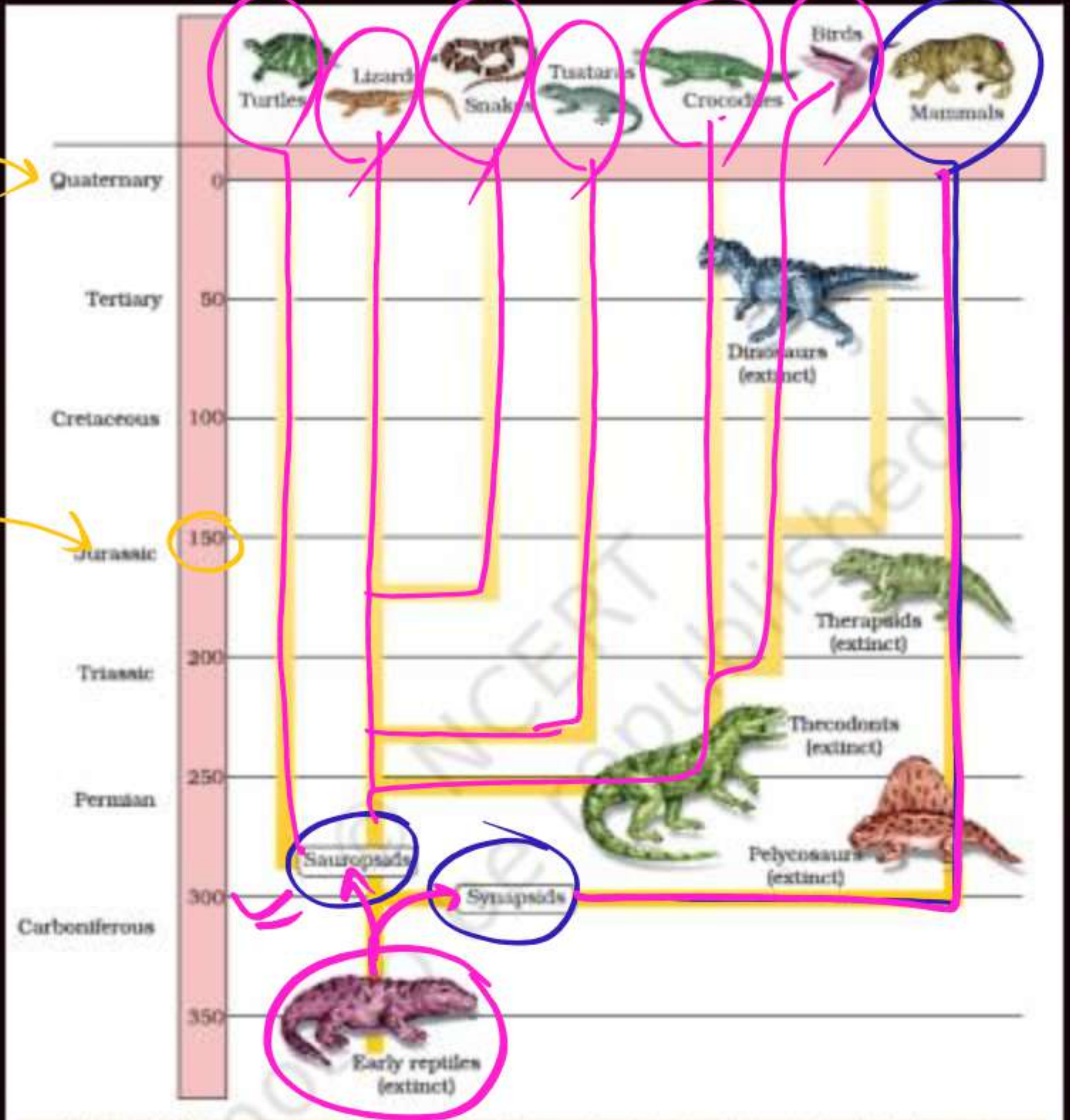


Figure 6.10 Representative evolutionary history of vertebrates through geological periods

Primats

- Prosimians → Lemur, Loris etc
- Anthropoids (Simians)
  - Monkeys
    - New World Monkey
      - eg - Spider monkey
    - Old World Monkey
      - eg - Rhesus monkey
      - [Closed to Human]
  - Apes
    - Gibbon (100 cc)
    - Orangutan (400 cc)
    - Gorilla (550 cc)
    - Chimpanzee (400 cc)
    - [Closed to Human]

Human

↓

1. Rama - Shiva pithacus
   ↓
2. Australopithecus (Ape - Man) (500 - 700 cc)
   ↓
3. Pre historic Man
   - Homo habilis (650 - 800)
   - Homo erectus (900 cc)
   ↓
4. Modern Man
   - Neanderthal (1400 cc)
   - Cromagnon (1650 cc)
   - Homo Sapiens - Sapiens (1400 cc)

15- M.Y.A → Dryopithecus = Ape like

3.4- M.Y.A → Ramapithecus = Man like → Ethopia & Tanzania → 4 feet

2-5- M.Y.A → Australopithecus = Ape-Man

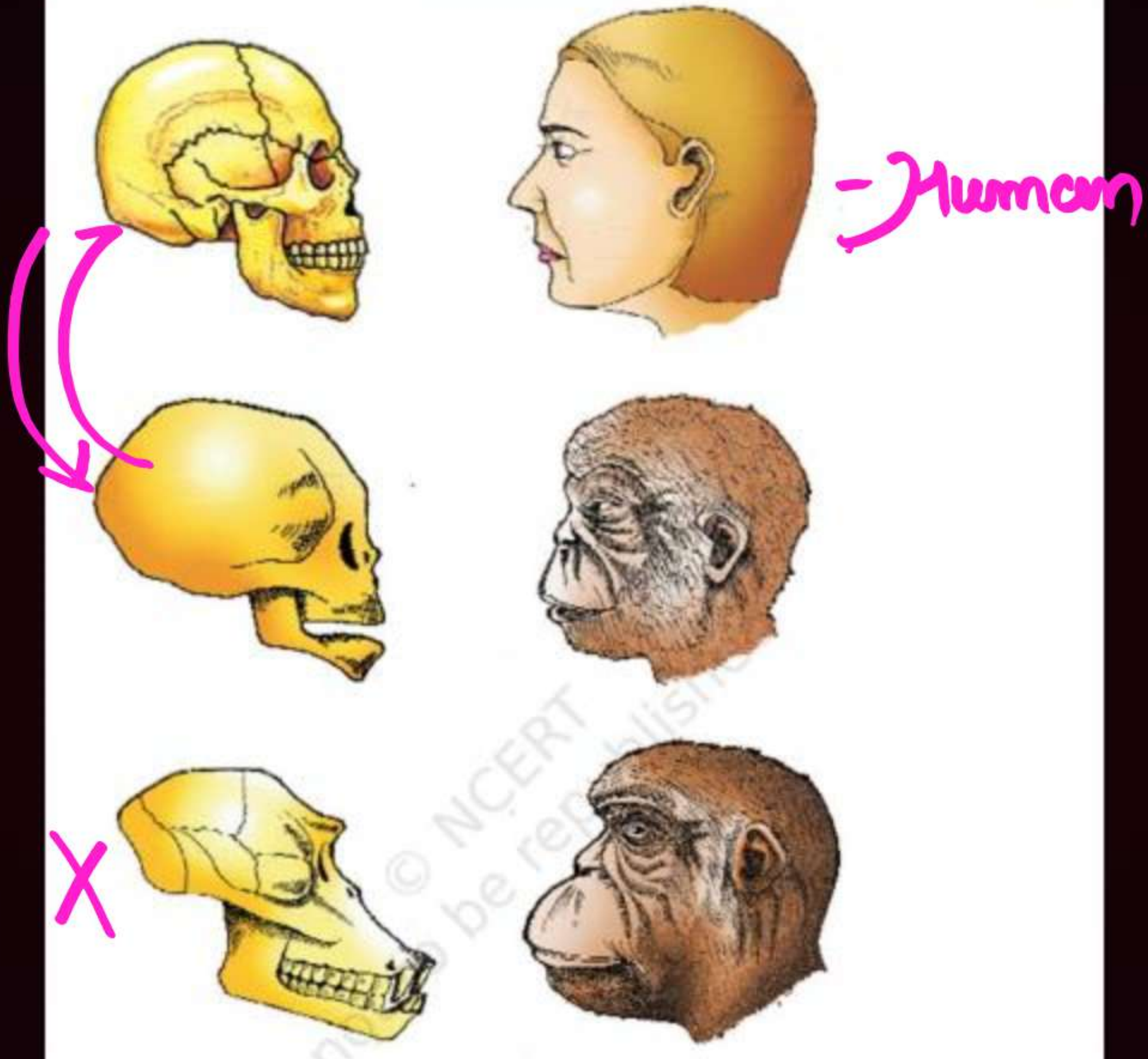
2-M.Y.A → Homo habilis = Skillfull Man - Not eat Meat

1.5 - M.Y.A → Homo erectus = Fire, eat Meat

10,0000-40000 Y.B → Neanderthal = Speech, Civilisation

50,000 - 10,000 Y.B → Cromagnon = Cave Panting

10,000 Y.B. → Homo-Sapiens-Sapi. = Agriculture



A comparison of the skulls of adult modern human being, baby chimpanzee and adult chimpanzee. The skull of baby chimpanzee is more like adult human skull than adult chimpanzee skull

Sweet potato and potato represent a certain type of evolution. Select the correct combination of terms to explain the evolution. **(NEET 2025)**

(a) Homology, convergent (b) Analogy, divergent

(c) Analogy, convergent (d) Homology, divergent

The flippers of the Penguins and Dolphins are the example of the (NEET 2024)

(a) Natural selection

(b) Convergent evolution

(c) Divergent evolution

(d) Adaptive radiation

Which of the following factors will not affect the Hardy-Weinberg equilibrium? (NEET 2024)

(a) Genetic drift

(b) Gene migration

(c) Constant gene pool

(d) Genetic recombination

Given below are some stages of human evolution. Arrange them in correct sequence. (Past to Recent)

A. *Homo habilis*

B. Homo sapiens

C. *Homo neanderthalensis*

D. *Homo erectus*

Choose the correct sequence of human evolution from the options given below: (NEET 2024)

(a) B-A-D-C

(b) C-B-D-A

(c) A-D-C-B

(d) D-A-C-B

Match List I with List II.

List I	List II
A. Mesozoic Era	I. Lower invertebrates
B. Proterozoic Era	II. Fish & Amphibia
C. Cenozoic Era	III. Birds & Reptiles
D. Paleozoic Era	IV. Mammals

Choose the correct answer from the options given below. (NEET 2024)

(a) A-III, B-I, C-II, D-IV

(b) A-I, B-II, C-IV, D-III

(c) A-III, B-I, C-IV, D-II

(d) A-II, B-I, C-III, D-IV

Select the correct group/set of Australian Marsupials exhibiting adaptive radiation. (NEET 2023)

(a) Numbat, Spotted cuscus, Flying phalanger

(b) Mole, Flying squirrel, Tasmanian tiger cat

(c) Lemur, Anteater, Wolf

(d) Tasmanian wolf, Bobcat, Marsupial mole

Natural selection where more individuals acquire a specific character value other than the mean character value, leads to: **(NEET 2022)**

(a) Stabilising change

(b) Directional change

(c) Disruptive change

(d) Random change

Which of the following statements is not true ?

(NEET 2022)

(a) Analogous structures are a result of convergent evolution

(b) Sweet potato and potato is an example of analogy

(c) Homology indicates common ancestry

(d) Flippers of penguins and dolphins are a pair of homologous organs

The factor that leads to Founder effect in a population is: (2021)

(a) Mutation
(b) Genetic drift
(c) Natural selection
(d) Genetic recombination

Inspite of interspecific competition in nature, which mechanism the competing species might have evolved for their survival? (2021)

(a) Mutualism
(b) Predation
(c) Resource partitioning
(d) Competitive release

Match List - I with List - II

	List – I		
(a)	Adaptive radiation	(i)	Selection of resistant varieties due to excessive use of herbicides and pesticides
(b)	Convergent Evolution	(ii)	Bones forelimb in Man and Whale
(c)	Divergent evolution	(iii)	Wings of Butterfly and Bird
(d)	Anthropogenic action	(iv)	Darwin Finches

(a) (a) – (ii), (b) – (i), (c) – (iv), (d) – (iii)

(b) (a) – (i), (b) – (iv), (c) – (iii), (d) – iii)

(c) (a) – (iv), (b) – (iii), (c) – (ii), (d) – (i)

(d) (a) – (iii), (b) – (ii), (c) – (i), (d) – (iv)

From his experiments, S.L. Miller produced amino acids by mixing the following in a closed flask:

(2020)

(a) CH_4, H_2, NH_3 and water vapor at 600 °C

(b) CH_3, H_2, NH_3 and water vapor at 600 °C

(c) CH_4, H_2, NH_3 and water vapor at 800 °C

(d) CH_3, H_2, NH_4 and water vapor at 800 °C

Flippers of Penguins and Dolphins are examples of:

(2020)

(a) Industrial melanism

(b) Natural selection

(c) Adaptive radiation

(d) Convergent evolution

Which of the following refer to the correct example(s) of organisms which have evolved due to changes of environment brought about by anthropogenic action?

(2020)

(i) Darwin's Finches of Galapagos islands.

(ii) Herbicide resistant weeds

(iii) Drug resistant eukaryotes.

(iv) Man-created breeds of domesticated animals like dogs.

(a) (ii),(iii) & (iv) (b) only (iv)

(c) only (i) (d) (i) & (iii)

Embryological support for evolution was disapproved by:

(2020)

(a) Charles Darwin (b) Oparin

(c) Karl Ernst von Baer (d) Alfred Wallace

Match list I with list II. (2020)

	List I		List II
(a	Adaptiveradiati on	(i)	Selection ofresistant varieties due to excessive use of herbicides and pesticides
(b	Convergent evolution	(ii	Bones of forelimbs in Man and Whale
(c	Divergent evolution	(ii	Wings of butterfly and birds
(d	Evolution by Anthropogenic c	(i	Darwin Finches

	action		

(a) (a)- (ii); (b) - (i); (c) - (iv);(d) - (iii)
(b) (a)- (i); (b) - (iv); (c) - (iii);(d) - (ii)
(c) (a)- (iv); (b) - (iii); (c) - (ii);(d) - (i)
(d) (a)- (iii); (b) - (ii); (c) - (i);(d) - (iv)

THANK YOU !!